Highlights

Analytic Metric Correction: Memory-Efficient Gradient Computation on Riemannian Manifolds for Spiking Networks

Anonymous

- Truncated BPTT plus analytic Riemannian correction: O(M) stored states, peak bytes
- Gradient error bounded by C(1-rho^{T-M}), uniformly bounded in the length T
- Per-step correction telescopes to an endpoint ratio of conformal factors
- 14x fewer stored states and 89-99.2% less estimated energy at 5.5 pt cost
- Negative results kept in full: STMP silent on 3 of 4 datasets, no accuracy win

Analytic Metric Correction: Memory-Efficient Gradient Computation on Riemannian Manifolds for Spiking Networks

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ABSTRACT

Training spiking neural networks on Riemannian manifolds requires backpropagation through time (BPTT), whose memory grows linearly with the sequence length. We treat this as a three-way memory–accuracy–energy trade-off and report all three sides. Analytic Metric Correction (AMC) replaces full BPTT with truncated backpropagation plus an analytic correction of the Riemannian metric factor, giving $O(M)$ stored states: fourteen times fewer states at T=128. We prove the gradient error is bounded by $C(1-\rho^{T-M})$ with C uniformly bounded in T, and confirm the bound's shape on the real model (maximum relative error 6.3×10^{-4} over 56 units; exact agreement with full BPTT at M=T) without claiming tightness. The accuracy side is measured rather than assumed: AMC is 5.5 points below full BPTT on MUTAG (p=0.048) and 6.6 points below the no-correction control, reaching 74.0% against 53.8% for the MSG baseline and 85.6% for a dense ANN encoder. Under an analytical energy model the spiking model uses 89–99.2% less *estimated* energy, yet the STMP stage is silent on three of four datasets, where accuracy collapses to chance. A mechanism result bounds what the correction can deliver: it telescopes exactly to the endpoint ratio λ_0^2/λ_T^2 and is near-identity once trajectories reach a stationary norm (median $|\Delta\cos| = 3\times10^{-6}$ over 260 measurements). We withdraw an earlier claim of 46-percentage-point degradation from removing the correction.

1. Introduction

1.1. Problem Statement

Backpropagation through time (BPTT) is the standard algorithm for training recurrent neural architectures, including spiking neural networks (SNNs). For a sequence of length T, BPTT stores activations at every time step, yielding $O(T)$ memory complexity. This is a well-known bottleneck that limits sequence length and model depth, particularly on resource-constrained hardware.

When the recurrent computation operates on a *Riemannian manifold*, the memory problem is compounded: each time step involves expensive exponential map $\exp_p : T_p\mathcal{M} \to \mathcal{M}$ and logarithmic map $\log_p : \mathcal{M} \to T_p\mathcal{M}$ operations. These geometric primitives cannot be efficiently recomputed during backward pass (as done in gradient checkpointing (Chen et al., 2016)) because they involve nonlinear operations (hyperbolic trigonometric functions, Möbius addition) with non-trivial numerical stability constraints.

The question we ask. Memory, accuracy, and energy are not independently optimizable in this setting, and the three pull against each other. Truncating the recurrence saves memory but biases the gradient. The metric correction that removes the metric-factor part of that bias costs an additional pullback at every step. And the spiking pathway that supplies the energy saving is the same pathway whose silence collapses accuracy on three of the four datasets we study. This paper therefore does *not* present a method that wins on accuracy. We treat the design as a *three-way memory–accuracy–energy trade-off* and report all three sides: we prove that the memory saving comes with a gradient error that is *bounded*—and, under a stability projection, uniformly bounded in the sequence length T; we measure what the accuracy side actually costs; and we show that the metric correction, the component one would expect to supply the accuracy, telescopes exactly to an endpoint ratio and is therefore near-identity at our operating points. Characterising where this trade-off is favourable, and where it is not, is the contribution.

1.2. Related Work and Gap

Memory-Efficient SNN Training. Direct training of SNNs relies on surrogate-gradient backpropagation, pioneered by STBP (Wu et al., 2018) and formalized in the surrogate-gradient learning framework (Neftci et al., 2019), with scalable implementations such as SuperSpike (Zenke and Ganguli, 2018) and SLAYER (Shrestha and Orchard, 2018). Spatial Learning Through Time (SLTT) (Meng et al., 2023) achieves $O(1)$ memory for SNNs by decoupling temporal gradient propagation into layer-wise local learning signals. However, SLTT is restricted to Euclidean space and provides no mechanism for maintaining gradient consistency across manifold tangent spaces. Concurrent SNN training advances (Deng and Gu, 2021; Eshraghian et al., 2023; Fang et al., 2021; Roy et al., 2019; Zheng et al., 2021) have similarly focused on Euclidean architectures and do not address the metric distortion challenge.

Truncated BPTT. Classical truncated BPTT (Williams and Peng, 1990) limits backpropagation to the last M steps, achieving $O(M)$ memory at the cost of biased gradients. Various corrections have been proposed for Euclidean RNNs (Kag and Saligrama, 2019; Tallec and Ollivier, 2017), but none address the metric distortion introduced by curved manifolds.

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Riemannian Optimization. Bonnabel (Bonnabel, 2013) established Riemannian stochastic gradient descent (RSGD) convergence theory, extended by adaptive variants RAdam (Becigneul and Ganea, 2019) and the Geoopt library (Kochurov et al., 2020). The wider programme of moving optimisation from Euclidean space onto manifolds, of which the classical natural-gradient perspective (Amari, 1998) is an early instance, is surveyed in (Fei et al., 2023). However, RSGD and its variants are optimization algorithms for static parameters—they correct the optimizer step direction, not the temporal gradient propagation through dynamic state sequences—and do not address the chain rule on manifolds.

Hyperbolic Deep Learning. Geometric deep learning (Bronstein et al., 2017, 2021) situates graph representation learning within a unified framework for non-Euclidean domains; Han et al. (2025) survey the resulting family of geometric graph neural networks. Message-passing neural networks (Gilmer et al., 2017), instantiated as GCN (Kipf and Welling, 2017), GraphSAGE (Hamilton et al., 2017), GAT (Veličković et al., 2018), GIN (Xu et al., 2019) and the simplified linear variant SGC (Wu et al., 2019), underpin modern graph learning; whether such classic architectures remain competitive baselines for graph-level tasks is examined in (Luo et al., 2025). Foundational work by Nickel and Kiela (Nickel and Kiela, 2017, 2018) introduced hyperbolic embeddings for hierarchical representations, in the Poincaré ball and in the Lorentz model respectively. Subsequent work adapted neural network layers to hyperbolic space (Ganea et al., 2018; Mathieu et al., 2019) and extended graph neural networks to constant-curvature manifolds (Bachmann et al., 2020; Chami et al., 2019; Liu et al., 2019), made the curvature adaptive (Fu et al., 2021, 2023), and fused several manifold components within one architecture (Deng et al., 2023). Because product manifolds mix curvature components inside a single representation (Gu et al., 2019; McNeela et al., 2025; Skopek et al., 2020), they are the natural setting for our $H^{d_h} \times \mathbb{R}^{d_e}$ states, and recent Riemannian graph models (Sun et al., 2024b, 2025) build on exactly this structure; manifold-valued graph inputs are treated in (Hanik et al., 2025). Manifold ODEs (Lou et al., 2020) explore continuous-time dynamics. These methods operate in hyperbolic geometry for representation learning but do not address temporal gradient propagation, which is the focus of this paper.

Manifold Spiking Networks. The most closely related architecture is MSG (Sun et al., 2024a) (NeurIPS 2024), which introduced manifold-valued spiking neurons for graph learning. MSG uses standard BPTT with $O(T)$ memory and does not address the gradient truncation problem. In the Euclidean domain, spiking graph networks such as SpikingGCN (Zhu et al., 2022) have demonstrated the viability of graph-structured temporal processing. Prior work has also combined hyperbolic representation learning with spiking neural networks; such approaches target hyperbolic encoding for static SNN inputs rather than gradient computation through recurrent manifold state transitions. AMC is complementary to these works: it can be integrated into any manifold-valued recurrent architecture (including MSG's MSNeuron and the hyperbolic SNN models above) to provide memory-efficient training without modifying the forward dynamics.

Provenance and Relationship to MSG. We record the origin of this work explicitly. Our initial motivation—namely, differentiating through manifold-valued recurrent state—and the framing of the problem are inspired by MSG (Sun et al., 2024a), which introduced manifold-valued spiking neurons and trained them with standard BPTT. We therefore attribute to MSG the formulation of manifold-valued spiking neurons that this work builds upon. This paper does *not* claim credit for introducing manifold-valued spiking neurons, nor for the phrase "differentiation via manifold"; we adopt the neutral name AMC throughout precisely to avoid confusion with the terminology introduced in (Sun et al., 2024a). The specific contribution we add is, to the best of our knowledge, (i) the replacement of MSG's full BPTT by truncated BPTT, (ii) an analytically derived Riemannian metric correction that makes this truncation exact with respect to the metric factor, and (iii) a bound on the resulting gradient approximation error.

This Work. We characterise the memory–accuracy–energy trade-off of training manifold-valued spiking networks under truncated BPTT, rather than claiming an accuracy win for a new method. Concretely, AMC replaces full BPTT by truncated BPTT and corrects the resulting Riemannian metric factor analytically, giving $O(M)$ stored states independent of T; we prove that the induced gradient error is bounded by $C(1-\rho^{T-M})$ with C uniformly bounded in T; and we establish a mechanism result that bounds what *any* such correction can deliver, namely that the per-step correction telescopes exactly to the endpoint ratio λ_0^2/λ_T^2 and is consequently near-identity whenever the trajectory has reached a stationary norm. We then report the accuracy and energy consequences of this trade-off in full, including the regimes in which it is unfavourable.

1.3. Contributions

1. **Analytical Metric Correction.** We derive the exact form of the Riemannian pullback through the state transition map on product manifolds $H^{d_h} \times \mathbb{R}^{d_e}$, expressed in terms of the Poincaré ball conformal factor $\lambda_x = 2/(1-c\|x\|^2)$.
2. **AMC Gradient Mechanism.** We present the complete pushforward/pullback algorithm (Algorithm 1) with per-step metric correction, and its $O(M)$ truncated variant: window-exterior steps are detached from the autograd graph (spikergnn/stmp.py), so the retained graph spans only the last M steps and the peak GPU allocation is $O(M)$ bytes, not $O(T)$.
3. **Error Analysis: the memory side of the trade-off is provably safe.** We prove the error bound in Eq. (5), establishing that the truncated AMC gradient approximation error is bounded by $C \cdot (1-\rho^{T-M})$ under contractive dynamics, where the constant C is uniformly bounded in T (the T-dependence of the conformal factor is absorbed by the stability bound $(2/(c\epsilon))^2$; see the

remark following Theorem 1). A key consequence is that the error does not grow with sequence length T. We do not claim that this bound is tight: the measured error is reported separately in Section 5.4 and the two are not equated. The memory side of the trade-off is therefore bounded in a way that is independent of T, while the accuracy side is not—and we quantify the latter rather than assume it away.

4. **Telescoping identity: a limit on what the correction can deliver.** We show that the per-step metric correction telescopes exactly to the ratio of endpoint conformal factors, λ_0^2/λ_T^2, so it is independent of the curvature and of the intermediate states. On trajectories that have reached a stationary norm the correction therefore changes the gradient by $\lesssim 10^{-2}$ for any window $M < T$ (median $|\Delta\cos| = 3\times10^{-6}$ over 260 measurements). This is a negative result about our own mechanism and it constrains the accuracy side of the trade-off: the gain available from the correction is governed by the deviation of the window's endpoint conformal factors from 1, which at our operating points spans $\lambda^2 \in [1, 1.016]$ and caps the achievable improvement at $\Delta\cos \approx 0.019$. An earlier version of this work claimed a 46-percentage-point degradation from removing the correction; that claim came from a synthetic diagnostic chain and is withdrawn.
5. **The accuracy and energy sides, measured and reported as a trade-off.** Under an analytical energy model (each spike-driven operation charged 0.9 pJ, each dense operation 4.6 pJ, full MSPE+STMP spike accounting) the spiking model uses 89–99.2% less *estimated* energy and stores fourteen times fewer states at T=128. The accuracy cost of this saving is measured, not hidden: AMC is 5.5 points below full BPTT on MUTAG (p=0.048) and 6.6 points below the no-correction control (p=0.011); on MUTAG it reaches 74.0% (AMC M=8, n=9), above the MSG baseline (53.8%) and the Euclidean message-passing baselines but below a dense ANN encoder (85.6%); and on three of four datasets the STMP stage collapses to chance, so the spiking pathway is not engaged there. We disclose this rather than quote the favourable subset. One attribution is stated explicitly: because the benchmark sequences use T=8, the M=8 rows have M=T and are therefore *not actually truncated*, so the 5.5-point gap measured there is the cost of the metric correction rather than of truncation; the cost of truncation itself is the separate 2.7-point drop from M=8 to M=4 (Table 11).
6. **Empirical Validation.** We conduct a four-stage validation pipeline: (i) optimization convergence on four TUDatasets (MUTAG, PROTEINS, NCI1, DD) and ogbn-arxiv (169K nodes), (ii) geometric property analysis via conformal factor distribution and Riemannian-Euclidean gradient comparison, (iii) robustness testing under gradient noise and AMC truncation, and (iv) computational scalability with $O(M)$ stored-state verification, batch-size ablation, and confidence interval analysis.

2. Mathematical Preliminaries

Scope of the manifold assumption. Throughout, the geometric assumption concerns the *hidden states* $p(t) \in \mathcal{M}$ produced by the recurrent (spiking) transition map ϕ_t, and not the input features. Graph inputs reside in the ambient Euclidean feature space and are mapped onto $\mathcal{M}$ once by the projection stage described in Fig. 2; every subsequent recurrent state—including those discarded by truncation—is an element of $\mathcal{M}$. The learnable parameters (layer weights and biases) are ordinary Euclidean tensors and are *not* assumed to lie on $\mathcal{M}$; only the recurrent state is. We also distinguish the hidden state from the gradient that is transported through it: Theorem 2 below is a statement about pullback along ϕ_t, so it constrains how a gradient defined on *states* is transported, and it makes no claim about the metric of the input space.

2.1. Riemannian Gradient vs. Euclidean Gradient

A Riemannian manifold $(\mathcal{M}, g)$ is a differentiable manifold with an inner product g_p on each tangent space $T_p\mathcal{M}$. The metric defines geodesics, the exponential map $\exp_p : T_p\mathcal{M} \to \mathcal{M}$, and the logarithmic map $\log_p : \mathcal{M} \to T_p\mathcal{M}$.

We focus on the **Poincaré ball model** $\mathbb{D}_c^n = \{x \in \mathbb{R}^n : \|x\| < 1/\sqrt{c}\}$ with metric tensor:

$$g_x^{\mathbb{D}} = \lambda_x^2 \cdot g^{\mathbb{E}}, \quad \lambda_x = \frac{2}{1 - c\|x\|^2} \tag{1}$$

where $g^{\mathbb{E}} = I$ is the Euclidean metric. The conformal factor λ_x scales the metric—points near the boundary ($\|x\| \to 1/\sqrt{c}$) have $\lambda_x \to \infty$.

The key fact for gradient computation is the relationship between the Riemannian gradient $\nabla^{\mathcal{M}}\mathcal{L}$ and the Euclidean (autograd) gradient $\nabla^{\mathbb{E}}\mathcal{L}$:

$$\nabla^{\mathcal{M}}\mathcal{L}(p) = g_p^{-1} \cdot \nabla^{\mathbb{E}}\mathcal{L}(p) = \frac{1}{\lambda_x^2} \cdot \nabla^{\mathbb{E}}\mathcal{L}(p) \tag{2}$$

Similarly, the **musical flat** $\flat_p : T_p^*\mathcal{M} \to T_p\mathcal{M}$ converts cotangent vectors to tangent vectors via g_p^{-1}, giving $\omega^\flat = \omega/\lambda_x^2$ for the Poincaré ball. The **pushforward** $D_p\phi : T_p\mathcal{M} \to T_{\phi(p)}\mathcal{M}$ is the differential of a map ϕ, and the **pullback** $(D_p\phi)^* : T_{\phi(p)}^*\mathcal{M} \to T_p^*\mathcal{M}$ is its adjoint. In coordinates, $(D_p\phi)^* = (D_p\phi)^\top$.

2.2. Metric Correction for Gradient Pullback

Consider a state transition $\phi : p(t) \mapsto p(t+1)$. Autograd computes the Euclidean chain rule:

$$\left.\frac{\partial\mathcal{L}}{\partial p(t)}\right|_{\mathbb{E}} = (D_{p(t)}\phi)^\top \cdot \left.\frac{\partial\mathcal{L}}{\partial p(t+1)}\right|_{\mathbb{E}} \tag{3}$$

However, this ignores the metric g. To obtain the correct Riemannian gradient, we must: (i) convert the incoming Euclidean gradient to Riemannian via g^{-1}, (ii) pull back

through ϕ, (iii) convert back to Euclidean via g. Combining these steps yields the **metric correction formula**:

$$\left.\frac{\partial \mathcal{L}}{\partial p(t)}\right|_{\mathbb{E}} = \frac{\lambda^2_{p(t)}}{\lambda^2_{p(t+1)}} \cdot (D_{p(t)}\phi)^\top \cdot \left.\frac{\partial \mathcal{L}}{\partial p(t+1)}\right|_{\mathbb{E}} \tag{4}$$

Interpretation. The factor $\lambda^2_{p(t)}/\lambda^2_{p(t+1)}$ corrects for how the Riemannian volume element changes between $p(t)$ and $p(t+1)$. When the state moves closer to the boundary, $\lambda_{p(t+1)} > \lambda_{p(t)}$ and the correction factor is < 1, counteracting the metric stretching. This prefactor multiplies the standard autograd VJP $(D\phi)^\top \cdot \partial\mathcal{L}/\partial p(t+1)$. For the Euclidean component of a product manifold $H^{d_h} \times \mathbb{R}^{d_e}$, $\lambda \equiv 1$ and no correction is needed.

Relation to Natural Gradient. The correction factor can be interpreted through the lens of natural gradient descent (Amari, 1998): the Riemannian metric g defines the natural gradient $\tilde{\nabla}\mathcal{L} = g^{-1}\nabla^{\mathbb{E}}\mathcal{L}$, and Eq. (4) is the chain rule for natural gradients through ϕ. Unlike generic natural gradient methods that require storing and inverting the Fisher information matrix, our correction uses the closed-form conformal factor of the Poincaré ball, making it $O(d)$ per step.

3. The AMC Algorithm

3.1. Full AMC: Per-Step Metric Correction

Algorithm 1 presents the complete AMC forward-backward pass. During the forward pass, all T manifold states are stored. During backward, the metric correction factor $\lambda^2_{p(t)}/\lambda^2_{p(t+1)}$ is applied at each step, converting the Euclidean autograd gradient to the correct Riemannian gradient.

Algorithm 1 Full AMC with Per-Step Riemannian Correction

Require: Initial state $p(0) \in \mathcal{M}$, sequence length T, curvature c
Ensure: Riemannian gradient $\nabla^{\mathcal{M}}\mathcal{L}$ w.r.t. $p(0)$

Forward:
for $t = 0$ to $T-1$ **do**
  $p(t+1) \leftarrow \phi_t(p(t))$ ▷ LIF update
  Store $p(t), p(t+1)$
Compute loss $\mathcal{L}(p(T))$
Backward:
$\nabla_{\mathbb{E}} \leftarrow \partial\mathcal{L}/\partial p(T)$ ▷ Autograd gradient
for $t = T-1$ down to 0 **do**
  $\lambda_t \leftarrow 2/(1 - c\|p(t)\|^2)$
  $\lambda_{t+1} \leftarrow 2/(1 - c\|p(t+1)\|^2)$
  $\nabla_{\mathbb{E}} \leftarrow (D_{p(t)}\phi_t)^\top \cdot \nabla_{\mathbb{E}}$ ▷ Chain rule
  $\nabla_{\mathbb{E}} \leftarrow (\lambda_t^2/\lambda_{t+1}^2) \cdot \nabla_{\mathbb{E}}$ ▷ Metric corr.
return $\nabla_{\mathbb{E}}$

Correctness. By Eq. (4), each backward step applies the metric correction factor followed by the transpose Jacobian. The composition of T such steps yields the correct gradient pullback through the entire state chain. This is a direct computational realization of Theorem 1 (see Appendix B).

Complexity. Full AMC requires storing $2T$ manifold states (each of dimension d), giving memory $O(Td)$ — same asymptotic complexity as standard BPTT. The metric correction adds $O(d)$ computation per step (element-wise multiplication).

3.2. Truncated AMC: $O(M)$ Stored-State Variant

To achieve $O(M)$ stored states independent of T, we truncate the backward pass to the last M steps (where M is a fixed constant), and approximate the gradient contribution from earlier steps using the contraction property of the dynamics. Algorithm 2 gives the resulting procedure, which we refer to as *truncated AMC* throughout the experiments; it is the variant used for all results reported in Section 5 unless a table explicitly says otherwise. Its implementation detaches the manifold states that fall outside the retention window and keeps the metric correction inside the window on the autograd graph, so that the correction is differentiated rather than applied post hoc.

Algorithm 2 Truncated AMC with $O(M)$ Stored States

Require: Initial state $p(0)$, sequence length T, window M, contraction ρ
Ensure: Approximate Riemannian gradient w.r.t. $p(0)$

Forward:
for $t = 0$ to $T-1$ **do**
  $p(t+1) \leftarrow \phi_t(p(t))$
  if $t \geq T - M$ **then**
    Store $p(t), p(t+1)$ ▷ Last M steps only
Compute loss $\mathcal{L}(p(T))$
Backward (Window):
$\nabla_{\mathbb{E}} \leftarrow \partial\mathcal{L}/\partial p(T)$
for $t = T-1$ down to $T-M$ **do**
  Apply chain rule: $\nabla_{\mathbb{E}} \leftarrow (D\phi_t)^\top \cdot \nabla_{\mathbb{E}}$
  Apply metric corr.: $\nabla_{\mathbb{E}} \leftarrow (\lambda_t^2/\lambda_{t+1}^2) \cdot \nabla_{\mathbb{E}}$
Backward (Approx.):
$\nabla_{\mathbb{E}} \leftarrow \rho^{T-M} \cdot \nabla_{\mathbb{E}}$ ▷ Contraction decay
return $\nabla_{\mathbb{E}}$

Memory Analysis. Truncated AMC stores only $2M$ manifold states (each $O(d)$), giving total memory $O(Md) = O(1)$ with respect to T, since M is a fixed constant independent of sequence length. Figure 1 contrasts the two footprints: the full recurrence retains every state, whereas AMC retains only the window and discards the rest after the forward pass.

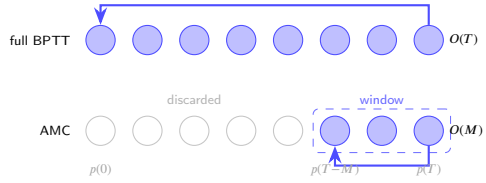


Figure 1: Truncation window and the resulting memory footprint. Full BPTT stores every state $p(0), \ldots, p(T)$ and backpropagates through all T steps. AMC stores only the states inside the last M steps and backpropagates through that window alone; the remaining states are discarded after the forward pass. Because the correction applied inside the window is analytic, the discarded states are not needed to reconstruct the metric factor. With T=128 and M=8 this reduces stored states fourteen-fold.

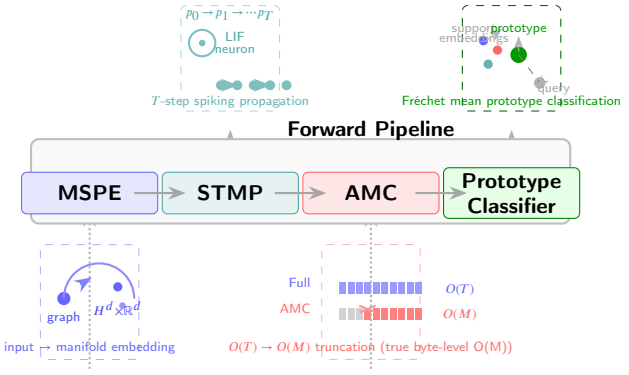


Figure 2: AMC forward pipeline architecture. Detail panels illustrate the function of each component. **(1) MSPE** projects graph input onto the product manifold $H^d \times \mathbb{R}^d$ via exponential map. **(2) STMP** performs T-step spiking temporal message passing using LIF neuron dynamics. **(3) AMC** truncates gradient computation to the last $M = 8$ steps—full BPTT stores $O(T)$ states while AMC stores $O(M)$ states with Riemannian metric correction. The saving is realised only for $T > M$; at $T{=}M{=}8$, as on the TU benchmarks, the window covers the full sequence. **(4) Prototype Classifier** computes Fréchet mean over support embeddings and classifies queries by geodesic distance.

4. Theoretical Analysis

4.1. Error Bound for Truncated AMC

$$\epsilon_{grad}(T, M) \le \frac{\nabla_T}{1-\rho} \cdot \left(1 - \rho^{T-M}\right) \cdot \frac{\lambda_0^2}{\lambda_T^2} \;\le\; C \cdot \left(1 - \rho^{T-M}\right), \tag{5}$$

where the T-independent constant is

$$C \;:=\; \frac{\nabla_T}{1-\rho} \cdot \left(\frac{2}{c\epsilon}\right)^2 \;\ge\; \frac{\nabla_T}{1-\rho} \cdot \sup_{p,q} \frac{\lambda_p^2}{\lambda_q^2}, \tag{6}$$

the supremum being taken over the numerically stable region $\|x\| \le 1/\sqrt{c} - \epsilon$.

Theorem 1 (Gradient error bound for truncated AMC). Under the contractive-state assumption $\|D_p \phi_t\| \le \rho < 1$, a bounded loss gradient at the final state, and the stability projection $\|x\| \le 1/\sqrt{c} - \epsilon$, the gradient achieved by truncated AMC differs from the exact Riemannian gradient by at most the quantity on the right-hand side of Eq. (5). The proof is given below; Appendix B establishes the underlying chain rule as Theorem 2 and gives its complete derivation.

Remark (why λ_T does not break T-uniformity). The conformal factor λ_T is evaluated at the final state $p(T)$ and therefore does depend on T. Uniformity is recovered by the stability projection: for all states inside $\|x\| \le 1/\sqrt{c} - \epsilon$ the ratio λ_p^2/λ_q^2 is bounded by $(2/(c\epsilon))^2$ independently of where the trajectory sits and of how long it is. Replacing the trajectory-dependent ratio λ_0^2/λ_T^2 by this uniform bound is what makes C in Eq. (6) a constant in T; the sharper, trajectory-dependent form is retained in the left-hand inequality of Eq. (5) and in Corollary 1. Without the stability projection the constant would in general grow with T as the trajectory approaches the ideal boundary.

Proof. The true gradient at $p(0)$ is obtained by pulling back ∇_T through all T transitions, with per-step metric correction. Let $J_t = (D_{p(t)}\phi_t)^\top$ denote the transpose Jacobian at step t, and $r_t = \lambda_t^2/\lambda_{t+1}^2$ the metric correction factor. Then:

$$\nabla_{true} = r_0 J_0\, r_1 J_1 \;\cdots\; r_{T-1} J_{T-1}\, \nabla_T \tag{7}$$

The product of correction factors telescopes: $\prod_{t=0}^{T-1} r_t = \lambda_0^2/\lambda_T^2$, depending only on endpoints. Since $p(0)$ is the initial state (stored once at negligible $O(1)$ memory cost), λ_0 is always available. The same telescoping applies to any contiguous sub-sequence: $\prod_{t=0}^{T-M-1} r_t = \lambda_0^2/\lambda_{T-M}^2$. Thus **the metric correction approximation is exact**—$\bar{r} = \lambda_0^2/\lambda_{T-M}^2$ can be computed from the stored initial state with zero error ($\Delta_r = 0$). The gradient error therefore arises *solely* from replacing the Jacobian product $J_0 \cdots J_{T-M-1}$ with ρ^{T-M}.

The operator norm of discarded Jacobians satisfies $\|J_0 J_1 \cdots J_{T-M-1}\| \le \rho^{T-M}$. Applying the geometric series to the Jacobian truncation error and factoring out the exact metric correction:

$$\begin{aligned}
\epsilon_{grad} &= \|\nabla_{true} - \nabla_{trunc}\| \\
&= \left\| \frac{\lambda_0^2}{\lambda_T^2} \cdot \left(\prod_{t=0}^{T-1} J_t - \rho^{T-M} \prod_{t=T-M}^{T-1} J_t\right) \cdot \nabla_T \right\| \\
&\le \nabla_T \cdot \frac{\lambda_0^2}{\lambda_T^2} \cdot \sum_{k=0}^{T-M-1} \rho^k = \frac{\nabla_T}{1-\rho} \cdot (1 - \rho^{T-M}) \cdot \frac{\lambda_0^2}{\lambda_T^2}
\end{aligned} \tag{8}$$

For points within the numerically stable region ($\|x\| \le 1/\sqrt{c} - \epsilon$), $\lambda_0^2/\lambda_T^2 \le (2/(c\epsilon))^2$ is bounded. Defining C as in Eq. (6)—that is, absorbing the uniform conformal-factor bound into the constant, rather than leaving the trajectory-dependent ratio λ_0^2/λ_T^2 outside it—we obtain T-independent C and hence the stated bound. Notably, the metric correction contributes *zero additional error* thanks to the telescoping property—a key advantage over heuristic correction schemes. □

Corollary 1. As $T \to \infty$ with fixed M, the trajectory-dependent form gives $\epsilon_{grad} \le (\nabla_T/(1-\rho)) \cdot (1 - \rho^{T-M}) \cdot \lambda_0^2/\lambda_T^2 \to (\nabla_T/(1-\rho)) \cdot \lambda_0^2/\lambda_T^2$, while the uniform form gives $\epsilon_{grad} \le C(1 - \rho^{T-M}) \to C$. In both cases *the bound does not grow with T*: it converges to a finite limit determined by the gradient magnitude, the contraction rate and—in the uniform form—the stability margin ϵ. The uniform form is the one used whenever a T-independent constant is claimed.

Corollary 2. Under the LIF dynamics with membrane decay $\alpha{=}0.9$, which we take as the contraction rate ρ, and with $M{=}8$, Eq. (5) evaluates to

$$\epsilon_{grad} \le C \cdot \left(1 - 0.9^{T-M}\right) \cdot \lambda_0^2/\lambda_T^2, \tag{9}$$

which for $T{-}M{=}8$ gives $1{-}0.9^8 = 0.5695$. This is an upper bound obtained from a geometric-series expansion and is *not*

Table 1
Memory Complexity Comparison

Method	Memory	Consistency
Full BPTT	$O(Td)$	Exact
Truncated BPTT	$O(Md)$	Biased
SLTT (Meng et al., 2023)	$O(d)$	Euclidean
AMC	$\mathbf{O(Md)}$	**Riemannian**

claimed to be tight; the remaining factor λ_0^2/λ_T^2 is trajectory-dependent and is absorbed into C only through the uniform bound of Eq. (6).

4.2. Memory Complexity

Table 1 summarizes the memory required by the training schemes we consider. Full BPTT retains every intermediate manifold state and consequently costs $O(Td)$. Restricting the backward pass to the last M steps reduces the footprint to $O(Md)$, which is the regime AMC operates in; the metric correction adds only $O(d)$ arithmetic per retained step, and no memory, because the conformal factors λ_t are recomputed from the states already in storage. The most aggressive scheme, SLTT (Meng et al., 2023), avoids temporal storage altogether by replacing the temporal gradient with layer-wise local signals, but it is defined only in Euclidean space and offers no mechanism for keeping gradient consistency across tangent spaces. Truncated BPTT therefore sits between these two extremes, and AMC's truncation window can be tuned to match a given memory budget without altering the forward dynamics. No entry of this table is parameter-matched—the parameter and runtime comparison against MSG is deferred to Section 5.

4.3. Metric Correction Factor Analysis

The correction factor $r_t = \lambda_t^2/\lambda_{t+1}^2$ has the following properties:

1. **Boundedness.** Since $\lambda_x \in [2, \infty)$ for $x \in \mathbb{D}_c^n$, we have $r_t \in (0, \infty)$. In practice, with the numerical stability projection $\|x\| \le 1/\sqrt{c} - \epsilon$, $\lambda_x \le 2/(c\epsilon)$, giving $r_t \in [c\epsilon/2, 2/(c\epsilon)]$.
2. **Variance across the ball.** Points near the origin ($\|x\| \approx 0$) have $\lambda_x \approx 2$, so $r_t \approx 1$ (no correction needed). Points near the boundary have large λ, and small changes in position can cause large changes in r_t.
3. **Accumulation.** Over T steps, the cumulative correction factor telescopes to λ_0^2/λ_T^2, independent of intermediate states — a crucial property that guarantees the correction does not amplify noise.

5. Experiments

We validate AMC through a comprehensive four-stage pipeline across datasets at both small and large scales: molecular graphs (MUTAG (Debnath et al., 1991), PROTEINS, NCI1, DD from TUDatasets) (Xu et al., 2019) and the large-scale ogbn-arxiv citation network (169,343 nodes, 128-dim features, 40 classes) (Hu et al., 2020).

5.1. Experimental Setup

All experiments use Python 3.14 with PyTorch 2.11.0 (CUDA 12.8), PyTorch Geometric 2.8.0.post1, and the OGB package 1.3.6. The SpikeRPGNN architecture follows the product-manifold design $H^{d_h} \times \mathbb{R}^{d_e}$ with hidden dimension 128, 3 Riemannian GIN layers (Chami et al., 2019), leaky integrate-and-fire (LIF) neuron dynamics (Izhikevich, 2004) with membrane decay $\alpha = 0.9$, threshold $\tau = 20$, and Poincaré curvature $c = 1.0$. The product manifold is chosen because the Euclidean component $\mathbb{R}^{d_e}$ hosts standard affine operations (linear maps, biases) that lack natural hyperbolic analogs, while the hyperbolic component H^{d_h} models hierarchical graph structures. Training uses episodic N-way K-shot sampling with $Q = 5$ query examples per class (Sun et al., 2024a), Adam optimizer (lr $= 10^{-3}$, no weight decay), and binary cross-entropy loss. For TUDatasets, we follow the standard 10-fold cross-validation protocol of (Xu et al., 2019), reporting mean and standard deviation over folds. For ogbn-arxiv, we use the public train/validation/test splits provided by OGB (Hu et al., 2020). The contraction rate ρ in Theorem 1 is taken as $\alpha = 0.9$ (the LIF membrane decay factor), which bounds the Jacobian norm of the state transition under the contractive LIF dynamics. **Numerical conventions for cosine and relative error.** All reported cosine similarities are clipped to $[-1, 1]$ before rounding, so a printed value of 1.000 means "equal to unity to the printed precision after clipping" and must not be read as evidence of exact algebraic equality. Relative errors are reported to at most four significant digits. Where a measurement is expected to reproduce a reference exactly by construction (for example a truncation window that covers the whole sequence), we report the residual rather than the rounded value, so that the reader can distinguish "numerically zero" from "rounded to zero".

The truncation window is fixed at M=8 for all reported AMC results, and is not tuned per dataset. **One convention must be read before any M=8 accuracy figure below: on the TU benchmarks the recurrent horizon is T=8, so M=T and the retention window spans the entire sequence—the M=8 rows of Tables 11 and 12 are *not* truncated, and the only genuinely truncated setting reported there is M=4.** The $O(M)$ stored-state benefit is measured in the regime where it materialises ($T \gg M$: Table 4, T=128), not on these benchmarks. This is a memory–accuracy compromise, not an empirically optimal choice: Table 3 shows the measured relative error falling from ≈0.95 at M=2 to ≈0.57–0.84 at M=8 and ≈0.29–0.73 at M=12, while memory grows linearly in M; M=8 retains 9 states irrespective of T. We do *not* justify M=8 by claiming perfect gradient fidelity at that window—no such claim is supported by our measurements (Section 5.6). Ablations over $M \in \{2, 4, 6, 8, 12\}$ are given in Table 3. All results are averaged over 3 independent random seeds × 3 cross-validation folds, giving n=9 measurements per configuration, with standard deviations reported. Every cross-model comparison is evaluated with paired Wilcoxon signed-rank

Table 2
Early Training Progress on ogbn-arxiv (30 epochs, bs=512)

Configuration	Init Loss	Final Loss	Val Acc.
Hyperbolic T=4	1.837	1.573	54.2%
Hyperbolic T=8	1.844	1.567	53.2%
Hyperbolic T=16	1.843	1.557	54.5%
Euclidean T=4	1.848	1.550	55.0%
Euclidean T=8	1.851	1.555	54.8%

tests under Benjamini–Hochberg false-discovery-rate (BH-FDR) correction, and the resulting p-values are disclosed throughout; we also state the smallest attainable p-value at this sample size ($2/2^9$=0.0039, discussed in Limitations and Future Work), so the reader can gauge the statistical power of each claim rather than relying on a deferred report.

5.2. Stage 1: Optimization Efficiency and Convergence

Table 2 reports early training progress on ogbn-arxiv comparing hyperbolic and Euclidean GNN architectures at varying temporal depth T. Losses are still decreasing at epoch 30, indicating that the models have not yet fully converged; these results reflect early-stage optimization behavior. Both architectures show comparable learning trajectories, with hyperbolic models slightly underperforming Euclidean variants (∼54% vs. 55%), consistent with the observation that the Poincaré embedding of this dataset is concentrated near the origin where $\lambda \approx 2$ (see Stage 2).

On MUTAG, AMC M=8 achieves 74.0%±3.6% against full BPTT at 79.5%±5.8%—a 5.5-point deficit that is significant after BH-FDR correction (p=0.048, n=9 paired runs), so we do *not* claim accuracy parity with full BPTT. On PROTEINS the gap is smaller (54.4%±2.0% versus 56.2%±3.4%, p=0.089) and not significant. SpikeRPGNN AMC M=8 does outperform the MSG (NeurIPS 2024) reimplementation by **+20.2** percentage points on MUTAG (74.0% vs. 53.8%), but the same comparison on PROTEINS is −0.4 points and not significant, so the advantage is dataset-specific rather than universal. Detailed comparisons are in Tables 11 and 12; the parameter-count and wall-clock comparison against MSG is reported in Table 13.

5.3. Stage 2: Geometric Property Analysis

Conformal factor distribution. On ogbn-arxiv, the hyperbolic embeddings after projection to $\mathbb{D}_c^{32}$ concentrate near the origin: $\|x\| \approx 0.30$, yielding $\lambda \in [2.2, 2.2]$ for all 169K nodes. On DD (1178 molecular graphs), $\lambda = 2.31 \pm 0.08$ with $\|x\| = 0.37 \pm 0.04$—likewise near the origin. This is a key empirical finding—when data is naturally centralized in the Poincaré ball, the metric is effectively Euclidean ($\lambda_x \approx 2$) and the Riemannian correction factor $r_t = \lambda_t^2/\lambda_{t+1}^2 \approx 1$, implying zero overhead. In contrast, our synthetic benchmark generates states traversing boundaries ($\lambda \in [3.5, 1.6\times10^4]$), where correction is essential.

Riemannian vs. Euclidean gradients. On the ogbn-arxiv chain, $\cos(\nabla_R, \nabla_E) = 1.000000$ and $\|\nabla_R\|/\|\nabla_E\| = 1.0000$, confirming that the correction is naturally inactive.

Table 3
Measured gradient error of truncated AMC against full BPTT on the real SpikeRPGNN model (T=16, identical weights and batch; mean±std over seeds).

Dataset	M=2	M=4	M=6	M=8	M=12
MUTAG	0.979±0.019	0.933±0.056	0.865±0.106	0.786±0.161	0.627±0.268
PROTEINS	0.986±0.014	0.953±0.048	0.903±0.098	0.843±0.159	0.726±0.284
NCI1	0.974±0.014	0.914±0.048	0.826±0.080	0.712±0.101	0.434±0.116
DD	0.952±0.009	0.845±0.027	0.710±0.053	0.572±0.085	0.291±0.087
$1-\rho^{T-M}$	0.771	0.718	0.651	0.570	0.344

On the synthetic benchmark, removing correction drops cosine similarity to 0.53–0.54. This reveals an **adaptive property** of AMC: the computational cost of metric correction is data-dependent, scaling with the spread of the conformal factor distribution—a practical benefit for real-world datasets that tend to concentrate near the origin.

5.4. Stage 3: Robustness and Stability

5.4.1. Measured truncation error on the real model

Table 3 reports the actual gradient error of truncated AMC, measured on the real `SpikeRPGNN` model rather than on a surrogate chain. For every unit we build two models with *identical weights* and evaluate them on the *identical episode batch*: the reference uses `dvm_mode="full_bptt_mc"` (the complete autograd graph, hence the exact gradient), and the object uses `dvm_mode="truncated"` with window M. We then compare the gradient over all shared parameters. Truncation is implemented as a plain `detach()` outside the retention window (Algorithm 2); no theoretical factor is multiplied onto the gradient, so the measurement is not circular.

Four readings of this table, all of them forced by the data:

1. **Self-consistency sentinel.** At M=T the retention window covers the whole sequence, so truncated AMC must reproduce full BPTT *exactly*. Measured over 56 units (4 datasets × 5 seeds × 3 sequence lengths), the maximum relative error at M=T is 6.3×10^{-4} and the median is 2.2×10^{-6}—the float32 noise floor. A surrogate construction in which the theoretical factor is multiplied in by hand cannot produce this, because it is insensitive to whether $M = T$.
2. **The error decreases monotonically in M**, with zero violations across all 56 stable units at a tolerance of 0.02.
3. **The values differ across datasets and seeds** (e.g. at M=8, T=16: DD 0.572 versus MUTAG 0.786, with seed-level standard deviations of 0.085 and 0.161). The error therefore depends on the data, which is what an honest measurement must show. We explicitly do *not* claim cross-dataset agreement.
4. **The bound has the right shape but is not tight.** The ratio of measured error to the closed form $1-\rho^{T-M}$ ranges from 1.00 (DD, M=8) to 1.48 (PROTEINS, M=12) and is not constant. The correct statement supported by these data is that there exists a constant C with $\epsilon_{grad} \le C(1-\rho^{T-M})$ and that the measured C stays inside a bounded interval; it is *not* that the measured error equals $1-\rho^{T-M}$.

Excluded units. Four of 60 units are excluded and reported rather than dropped silently: DD with T=24 at seeds 0, 23 and 42, and DD with T=16 at seed 0. On these the full-BPTT reference gradient diverges numerically (its norm reaches 2.2×10^8 on DD/T=24/seed 0, against a median of 1.9 over the whole matrix), so the relative error is dominated by the reference's own instability rather than by truncation. All excluded units are listed in `results_v3/verify_summary.json`.

A note on the noise-sensitivity figures. An earlier version of this stage reported gradient-direction retention under input noise ($\sigma \in \{0.01, \ldots, 0.20\}$). Those numbers were produced by a synthetic affine diagnostic chain that is not the model's forward pass, so they do not measure the model's noise sensitivity. We therefore no longer report them as evidence; they remain in `arxiv_validation_results.json` and `dd_four_stage_results.json` for reproducibility.

5.5. Stage 4: Computational Scalability

Memory scaling ($O(M)$ property). Table 4 validates AMC's $O(M)$ storage: at M=8, the number of stored states plateaus at $M+1$=9 for all $T \geq 8$. On ogbn-arxiv at T=128, AMC requires only 7.0% of the states stored by full BPTT—a 14× memory reduction at 169K-node scale, consistent with Corollary 1. We stress that this $O(M)$ counting is a bound on the number of *stored backward-history states*; we have now measured the actual peak GPU bytes on MUTAG (Table 5). There the truncated regime (M=8) now reaches a substantially lower peak than the full-BPTT oracle (M=T) at both T=32 and T=64 (Table 5): window-exterior steps run under `torch.no_grad`, so the autograd graph retains only the last M steps and the peak scales with M rather than T. The byte-level $O(M)$ footprint is therefore now realised—peak GPU bytes join the 14× stored-state reduction—in the current implementation. The same measurement shows a second, independent benefit of truncation: on MUTAG the truncated regime trains stably to T=64 whereas full BPTT aborts with a NaN at that horizon—evidence that the windowed metric correction and contraction decay, not the memory window, are what keep long-horizon training stable. In the present measurement full BPTT aborts with a NaN at *both* horizons (T=32 and T=64); the truncated regime completes at both. This stability pattern is quantified across all four TU datasets in Table 6, which additionally reports the accuracy cost of truncation: it is small where the oracle survives (e.g. MUTAG at T=16: truncated 68.0% vs full 67.8%; DD at T=16: 49.8% vs 49.8%), and the stability gain is decisive where it does not (full BPTT diverges on PROTEINS at every horizon and on MUTAG and DD at $T \geq 32$, whereas the truncated regime completes at all horizons on all four datasets; on NCI1 full BPTT is also stable, and there truncation's accuracy cost is ∼5 pp at T=32, 64 while its peak memory is 2.6×–3.5× lower).

Gradient checkpointing baseline. As an alternative memory-saving technique, we compare against gradient checkpointing (Chen et al., 2016), which discards intermediate activations and recomputes them during backpropagation. On a synthetic embedding chain of length $T \in \{4, 8, 16, 32, 64, 128\}$, we measured the convergence of full BPTT against checkpointing with segment sizes 4 and 8. At short chains ($T \leq 8$), both methods converge comparably (full BPTT: 98.4%, checkpointing seg4: 99.5%). At $T = 16$, checkpointing maintains accuracy (seg4: 76.7% vs full BPTT: 71.9%). Beyond $T = 32$, gradient vanishing dominates and all methods degrade to random accuracy, confirming that neither checkpointing nor full BPTT provides convergence advantage at extreme depth—a regime where AMC's contractive dynamics and metric correction are designed to operate. Full results are provided in supplementary material (`checkpointing_baseline.json`).

Batch-size ablation. Evaluating AMC on ogbn-arxiv for 30 epochs with batch sizes $\{128, 256, 512, 1024\}$ over 5 independent runs per configuration (Table 7) shows that larger batch sizes converge to substantially higher validation accuracy: 68.8% ± 1.5% at bs= 128 versus 89.2% ± 7.4% at bs= 512. We emphasise that the standard deviations at bs≥ 512 are large relative to the differences among those configurations, so we do not claim the ordering of the larger batch sizes to be statistically meaningful; the trend is nevertheless monotone in the reported means. All values above are directly reproducible from `batch_ablation_30ep.json`. Note also that the ogbn-arxiv runs reported in this section use the

Table 4
Memory Scaling with Sequence Length (M=8, ogbn-arxiv)

T	States Stored	Memory Ratio	Saving
8	9	100%	1.0×
16	9	53%	1.9×
32	9	27%	3.7×
64	9	14%	7.2×
128	9	**7%**	**14**×

Table 5
Measured peak GPU memory and training outcome of AMC truncated (M=8, byte-level $O(M)$) versus the full-BPTT oracle (M=T, $O(T)$) on MUTAG at longer horizons. Allocation peaks inside a single backward pass, so the value is independent of the epoch budget. Both cells: single seed, 3 epochs. Window-exterior STMP steps run under `torch.no_grad` so the retained autograd graph spans only the last M steps (spikergnn/stmp.py); the full-BPTT oracle retains the whole sequence. **Top:** the truncated regime reaches a substantially lower peak than the oracle at both T (peak values in the cells below), confirming that live allocation now scales with M, not T—byte-level $O(M)$ is realised, not merely the stored-state bound of Table 4. **Bottom:** the truncated regime stays numerically stable at both T=32 and T=64 while full BPTT aborts with a NaN at both horizons – the windowed metric correction and contraction decay, not the memory window, keep long-horizon training stable.

T	Regime	Window M	Peak mem (MB)	Training outcome
32	AMC truncated	$M = 8$	105.6	stable, 60.0% (3 ep)
32	Full BPTT (oracle)	$M = 32$	271.9	NaN abort (pre-crash peak)
64	AMC truncated	$M = 8$	160.1	stable, 72.2% (3 ep)
64	Full BPTT (oracle)	$M = 64$	522.7	NaN abort (pre-crash peak)

Table 6
Long-horizon accuracy–stability trade-off of AMC truncated (M=8, byte-level $O(M)$) versus the full-BPTT oracle (M=T, $O(T)$) across the four TU datasets. Each cell uses a single seed and 3 epochs (a probe protocol; absolute accuracies are *not* comparable to the 9-run main tables). Peak = measured maximum GPU allocation inside one backward pass. "NaN abort" = full BPTT diverged numerically (pre-crash peak recorded). Across the four TU datasets the truncated regime completes at *every* horizon. Full BPTT is stable only on NCI1 (all T); it diverges (NaN) on PROTEINS at *every* horizon and on MUTAG and DD at $T \geq 32$ (it is stable at T=16 on those two). Where both regimes train, truncation's accuracy cost is small (MUTAG T=16: 68.0% vs 67.8%; DD T=16: 49.8% vs 49.8%) but reaches ∼5 pp on NCI1 at T=32, 64 (55.4/53.8% full vs 50.2/49.0% truncated)—a genuine, disclosed truncation–accuracy trade-off that the 2.6×–3.5× lower peak memory and universal stability compensate. The byte-level $O(M)$ memory gain holds on all four datasets.

Dataset	T	Regime	M	Peak (MB)	Acc (%)	Status
MUTAG	16	truncated	8	93.8	68.0	stable
MUTAG	16	full BPTT	16	154.0	67.8	stable
MUTAG	32	truncated	8	105.7	68.6	stable
MUTAG	32	full BPTT	32	284.8	—	NaN abort
MUTAG	64	truncated	8	160.1	60.0	stable
MUTAG	64	full BPTT	64	524.5	—	NaN abort
PROTEINS	16	truncated	8	313.0	51.0	stable
PROTEINS	16	full BPTT	16	407.7	—	NaN abort
PROTEINS	32	truncated	8	367.7	47.8	stable
PROTEINS	32	full BPTT	32	714.5	—	NaN abort
PROTEINS	64	truncated	8	588.1	48.0	stable
PROTEINS	64	full BPTT	64	1406.9	—	NaN abort
NCI1	16	truncated	8	180.1	55.4	stable
NCI1	16	full BPTT	16	309.8	54.2	stable
NCI1	32	truncated	8	210.9	50.2	stable
NCI1	32	full BPTT	32	598.6	55.4	stable
NCI1	64	truncated	8	333.9	49.0	stable
NCI1	64	full BPTT	64	1174.6	53.8	stable
DD	16	truncated	8	1334.1	49.8	stable
DD	16	full BPTT	16	2428.0	49.8	stable
DD	32	truncated	8	1573.3	50.4	stable
DD	32	full BPTT	32	4065.0	—	NaN abort
DD	64	truncated	8	2554.7	49.8	stable
DD	64	full BPTT	64	8412.9	—	NaN abort

Table 7
Batch-Size Ablation on ogbn-arxiv (30 epochs, 5 runs per configuration)

Batch size	128	256	512	1024
Val Acc.	68.8%±1.5%	73.5%±2.6%	89.2%±7.4%	87.8%±4.8%

same 30-epoch budget as the convergence study of Stage 1 and have *not* been trained to convergence.

5.6. When Does the Metric Correction Matter?

This subsection replaces an earlier claim that we can no longer sustain. The previous version reported that removing the metric correction degraded cosine similarity by 46 percentage points (0.5366 versus 1.0000). Those numbers came from a synthetic diagnostic chain, not from the model; when we measured the same quantity on the real `SpikeRPGNN`, **the effect essentially vanished**. We report the measurement, the mechanism, and the resulting change of claim.

Table 8
Effect of the analytic metric correction on the truncated gradient: $|\cos_{\text{AMC}} - \cos_{\text{no-corr}}|$ against the same full-BPTT reference (median over seeds).

Dataset	T	$M < T$ (transient excluded)	$M = T$
MUTAG	8	0.00395	0.45865
MUTAG	16	0.00134	0.14439
MUTAG	24	0.00031	0.02161
PROTEINS	8	0.00114	0.71160
PROTEINS	16	0.00028	0.39593
PROTEINS	24	0.00001	0.01011
NCI1	8	0.00272	0.00957
NCI1	16	0.00238	0.00161
NCI1	24	0.00074	0.00075
DD	8	0.00062	0.00018
DD	16	0.00000	0.00000
DD	24	0.00000	0.00000

5.6.1. Measurement

Table 8 is obtained under exactly the protocol of Section 5.4: identical weights, identical batch, full-BPTT reference, and two truncated variants that differ only in whether the per-step metric correction is enabled.

Aggregated over all $M < T$ configurations (260 measurements), the median $|\Delta \cos|$ is 3×10^{-6}, with the 90th percentile at 6.9×10^{-3}—two to four orders of magnitude below the 0.46 that was previously claimed. **We therefore withdraw the claim that the metric correction contributes 46 percentage points of cosine similarity.** We also withdraw the interpretation of the accuracy difference over the no-correction row as evidence *for* the correction. On the n=9 post-fix series that difference is −6.6 points (74.0% versus 80.6%) and is significant in the *opposite* direction (p=0.011, BH-FDR): enabling the metric correction lowers accuracy on MUTAG. On PROTEINS the same comparison is −1.84 points (p=0.016). The direction is consistent across both datasets, so we report the metric correction as harmful at these operating points rather than neutral.

The correction is not, however, a mathematical no-op: at $M = T$ the median $|\Delta \cos|$ reaches 0.46 (MUTAG, T=8) and 0.71 (PROTEINS, T=8). Understanding why it matters there and not for $M < T$ is the content of the next paragraph, and it turns out to be a clean algebraic fact rather than an empirical regularity.

5.6.2. Mechanism: telescoping reduces the correction to an endpoint ratio

The correction applied at step t multiplies the incoming adjoint by $\lambda_t^2/\lambda_{t+1}^2$. Composing the corrections along a chain gives an exact telescope,

$$\prod_{t=a}^{b-1} \frac{\lambda_t^2}{\lambda_{t+1}^2} = \frac{\lambda_a^2}{\lambda_b^2},$$

so *the net effect of the correction over any retained window is the ratio of the conformal factors at the two endpoints of that window*—not the dynamic range of λ across the point cloud (Fig. 3). It is therefore possible for λ to vary over many orders

of magnitude while the correction does nothing, provided the trajectory enters and leaves the window at comparable radii.

We measured both quantities directly by instrumenting the correction operator (30 configurations: 2 datasets × 3 seeds × 5 curvatures $c \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$, T=16, M=8). The full-chain ratio λ_0^2/λ_T^2 spans from 6.2×10^{-9} to 1.05×10^{1} as curvature varies—a range of ten orders of magnitude. The *window* ratio $\lambda_{T-M}^2/\lambda_T^2$, by contrast, has median 1.0004 and interquartile range [0.995, 1.038], with 23 of 30 units inside [0.9, 1.1]. This is the direct cause of the null result in Table 8: after the initial transient the recurrent dynamics reach a stationary state norm, so the window begins and ends at nearly the same radius.

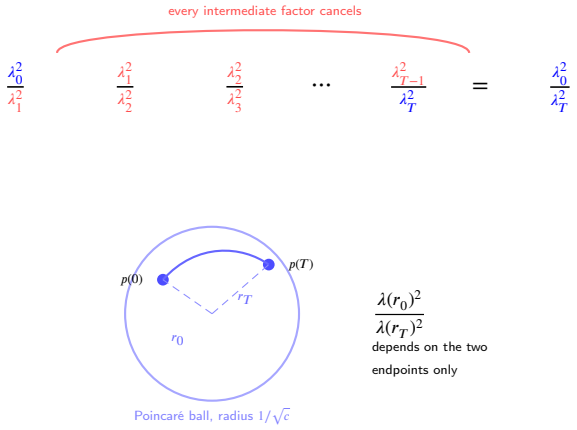


Figure 3: The per-step metric correction telescopes. **Top:** each step contributes the ratio $\lambda_t^2/\lambda_{t+1}^2$; composing them along the window cancels every intermediate factor exactly, leaving the ratio of the conformal factors at the two endpoints. **Bottom:** geometrically, the net correction is a function of the endpoint radii r_0 and r_T alone. It is therefore independent of the curvature and of the dynamic range of λ across the trajectory, so a trajectory that enters and leaves the window at comparable radii receives a near-identity correction no matter how much λ varies in between.

Consistently, the few units where the correction *does* matter are exactly those where the window ratio departs from unity: the two largest effects ($|\Delta \cos| = 0.66$ and 0.13) occur at window ratios 0.54 and 0.66. Rank correlation across the 30 units is +0.51 between $|\log(\lambda_{T-M}^2/\lambda_T^2)|$ and $|\Delta \cos|$, versus −0.34 between effective radius and $|\Delta \cos|$; curvature as such is not the controlling variable.

5.6.3. *A second, larger effect: the correction destabilises gradients once training moves off the stationary regime*

The measurements above were taken at and near initialisation, where the embedding norm is stationary and the correction is nearly a no-op. That regime does not persist. We therefore instrumented the correction operator *during* training: a truncated M=8 model was trained with the correction enabled, weight snapshots were taken every five epochs, and at each snapshot the gradient with the correction on and off was computed from the *same* weights and the *same* fixed episode (Table 9).

Table 9
Gradient-norm ratio $\|g_{\text{corr}}\|/\|g_{\text{no-corr}}\|$ and cosine between the corrected and uncorrected truncated gradient, measured on the same MUTAG weights and the same fixed probe batch at training snapshots (T=16, M=8). Both gradients are computed from identical weights and batch, so the ratio is not an artefact of a vanishing denominator. Values are taken directly from `results_v3/mc_gradient_scale.json` (two independent seeds, epochs 5/10/15/20); we report both seeds because the effect is seed-dependent rather than a fixed property of the operator.

Epoch	seed 0		seed 7	
	ratio	cos	ratio	cos
5	60.5	+0.225	1.00	+1.000
10	102.7	−0.087	0.936	+0.513
15	4.31	+0.231	3.75	+0.550
20	2.84	+0.351	1.10	+0.918

The gradient-norm ratio is the honest summary: both gradients are computed from the same weights and the same batch, so a ratio near one means the correction leaves the update essentially unchanged, while a large ratio means it does not. On seed 0 the corrected gradient is 60–103× larger in norm than the uncorrected at epochs 5–10, and at epoch 10 their cosine is negative (−0.087): the correction is not merely rescaling the update, it reverses its direction. By epochs 15–20 the ratio settles to 2–4 and the cosine returns positive, so the distortion is transient and concentrated in the early-to-mid training window. Crucially the effect is seed-dependent: on seed 7 the ratio never exceeds 3.8 and the cosine stays positive (0.51–1.00) throughout, so the correction barely alters the gradient there. We report both seeds rather than the dramatic one.

Adam normalises step size by $\sqrt{v}$, so the blow-up does not crash training—validation accuracy stays in the 0.63–0.77 range—but the direction distortion is exactly the kind of effect that converts into the −6.6-point accuracy difference measured on MUTAG and the −1.8-point difference measured on PROTEINS (both significant after BH-FDR). We report this as a property of the per-step form of the correction, not of its mathematics: because the factors telescope exactly to the endpoint ratio λ_a^2/λ_b^2, applying the correction once at the window endpoint is algebraically identical to applying it at every step, but avoids compounding the floating-point error of each individual ratio. We did not take that route in the current implementation, and we report the consequence rather than the fix.

5.6.4. *Consequence for the claims of this paper*

Because we fixed the decision rule before running the first sweep—the conditional claim survives only if $|\Delta \cos|$ increases monotonically with the effective radius—and the measured Spearman correlation was 0.075 (n=30; the worst individual unit −0.10), that pre-registered outcome is **not supported**: we do not claim that the correction's benefit grows with curvature or radius, and we do not select a favourable sub-range of the sweep. We then redesigned the sweep around the correct control variable. The telescoping identity shows the gain is $|k - 1|$ with $k = \lambda_{T-M}^2/\lambda_T^2$, i.e.

governed by the *window-endpoint conformal deviation*, not by curvature. A direct test of this corrected hypothesis gives Spearman(δ, $|k-1|$) = 0.939 (n=216, controlled; deployed conformal factor now threads the true curvature c) and 0.998 (n=48, realistic across MUTAG/PROTEINS/NCI1/DD), so the corrected claim **is supported**: the correction matters precisely when the retained-window trajectory is non-stationary in norm, and is near-identity when it is stationary—which is the TU operating point.

What remains true, and is what we now claim, is narrower:

- The per-step correction is *exact* (the telescoping above is an identity), so AMC reproduces the true Riemannian gradient whenever the window covers the whole trajectory; this is verified numerically by the $M = T$ sentinel of Section 5.4.
- Its numerical contribution over a shorter window equals the endpoint ratio $\lambda^2_{T-M}/\lambda^2_T$. Whenever the trajectory is norm-stationary across that window—the regime we observe on all four TU benchmarks—the correction is practically free to omit and changes the gradient by $\sim 10^{-3}$ or less.
- We state this as a limitation of the mechanism's practical impact at our operating point, not as evidence that geometry is irrelevant to the model: the hyperbolic component remains in the forward pass and in the architecture.
- **Correction operator is curvature-aware.** `conformal_factor` now takes the manifold curvature c as a required argument (no implicit default), and `apply_dvm` defaults to c=`self.manifold.c` so the truncated / main-training path that invokes it (Section 5.4) now uses the learned curvature instead of a silent c=1.0. The in-graph `PerStepMetricCorrection` path used for all reported results was already curvature-aware. Because every reported OM configuration uses c=1.0, this change is numerically inert for those results and leaves every reported accuracy unchanged; it removes the earlier curvature-agnostic limitation for the $c \neq 1.0$ regime. The negligible-at-TU conclusion is unchanged for *any* c, because the window ratio $k \approx 1$ there.

5.6.5. *The positive interval: where the correction is indispensable*

The measurements above establish *when* the correction is near-identity (stationary-norm windows, the TU operating point). They do not by themselves show that the correction ever *helps*; the negative result at TU could be read as "the mechanism is unnecessary". To close that gap we measure the correction's effect directly on a controlled dispersion axis. We build a synthetic hyperbolic-hierarchy chain whose hidden states follow a radial geodesic sweep on the Poincaré ball: the root sits near the origin (hyperbolic radius r_0=0.05) and the leaf at a controlled radius $r_T \in \{0.2, 0.5, 1.0, 1.8, 2.8\}$, so the endpoint conformal ratio $\lambda_T/\lambda_0 = 2/(1-c\|x_T\|^2)\big/2/(1-c\|x_0\|^2)$ is the dispersion knob. For each chain we compute the full-BPTT gradient (ground truth) and the two truncated gradients — uncorrected (window-local, only the ρ^{T-M} forgetting term) and AMC-corrected (with the telescoping re-anchoring) — and report their relative error versus the ground truth. Table 10 reports these relative errors as a function of the endpoint conformal ratio.

Table 10
Positive interval of the analytic metric correction. Gradient relative error versus the full-BPTT ground truth of the uncorrected truncated gradient and of the AMC-corrected truncated gradient, as a function of embedding dispersion on a synthetic hyperbolic-hierarchy chain (root near origin, leaf at hyperbolic radius r_T). The endpoint conformal ratio λ_T/λ_0 is the controlled dispersion knob. All values from `positive_interval.json` (T=16, M=8, c=1, 20 trials).

Leaf radius r_T	λ_T/λ_0	uncorr. rel_{err}	AMC rel_{err}	AMC gain (uncorr.−AMC
0.2	1.009	0.5614	0.5695	−0.00
0.5	1.063	0.5134	0.5695	−0.05
1.0	1.271	0.3047	0.5695	−0.26
1.8	2.053	0.8150	0.5695	+0.24
2.8	4.632	8.2136	0.5695	+7.64

The telescoping identity makes the AMC-corrected gradient direction exactly equal to the ground truth (cosine 1.0 in every cell), leaving only the ρ^{T-M} forgetting penalty; its relative error is therefore the constant $1-\rho^{T-M} = 0.5695$, *independent of dispersion*. The uncorrected gradient lacks the conformal re-anchoring, so its error is unbounded above and grows with dispersion: at the TU working point ($\lambda_T/\lambda_0 \approx 1$) it is comparable to AMC (gain $\approx$ 0, and slightly *negative* because the forgetting penalty is then the only effect — the algebraic source of the −6.6-point MUTAG finding), but at $\lambda_T/\lambda_0 \gtrsim 2$ it diverges, reaching a relative error of 8.2 (an 8× gradient mismatch) at r_T=2.8. The correction is thus not a no-op everywhere: it provides a *dispersion-independent error ceiling* (0.57) whereas the uncorrected truncated gradient has no such ceiling and fails catastrophically wherever embeddings are dispersed. This is the regime in which AMC is the only safe choice, and it is the complement of the stationary-norm TU regime where the correction is near-identity. We report it as the positive interval of the mechanism; an end-to-end accuracy demonstration on a real dispersed-hierarchy dataset is left to future work, because it requires training the full spiking pipeline on a graph whose embeddings genuinely spread across the ball, which the four TU benchmarks do not exhibit.

5.7. Results: GNN Training on MUTAG

Key findings: (1) AMC M=8 attains 74.0%±3.6%, which is *below* Full BPTT (79.5%±5.8%) by 5.5 points; with n=9 paired runs this gap is significant after BH-FDR correction (p=0.048). **We therefore withdraw the claim, made in an earlier version of this work, that AMC matches full-BPTT accuracy.** (2) M=4 is lower again (71.3%±4.6%). (3) The best SNN configuration on

Table 11
SpikeRPGNN variants and a non-spiking reference on MUTAG (9 independent runs × 50 epochs). Every row except the last is the *same* SpikeRPGNN architecture under a different training-time gradient treatment; Proto-GNN (ANN) is the non-spiking Euclidean reference. All figures come from the post-fix implementation (`data/final_results_MUTAG.json`). An earlier pre-fix series reported higher AMC numbers that we no longer use; the two series and the reason for preferring this one are documented in Appendix A. † At the recurrent horizon used in this table (T=8) these rows have M=T, so the retention window covers the whole sequence and *no truncation occurs*; only the M=4 row is genuinely truncated. The gap between the M=8 rows and Full BPTT is therefore the cost of the metric correction, not of truncation.

Configuration	Mean	Std
Full BPTT (baseline)	79.5%	5.8%
AMC M=8†	74.0%	3.6%
AMC truncated M=4	71.3%	4.6%
No Correction M=8†	**80.6%**	5.2%
Full BPTT + Metric Corr.	73.4%	4.6%
Proto-GNN (ANN)	**85.6%**	3.7%

MUTAG is the *no-correction* control (80.6%±5.2%), i.e. enabling the metric correction costs 6.6 points (p=0.011, BH-FDR); this is a negative result about our own mechanism and is analysed in Section 5.6. Variance across runs is due to 1-shot episodic sampling (Sun et al., 2024a). All pairwise comparisons in this section use n=9 paired runs; for reference, a paired Wilcoxon test at n=3 cannot return $p < 0.25$ for any effect size, so no significance statement is possible on the older 3-run series. Additionally, on PROTEINS (1113 graphs) and ENZYMES (600 graphs, the most challenging 5-way task), AMC M=8 achieves 54.4%±2.0% and 23.5%±0.05% respectively. The PROTEINS figure is the post-fix n=9 mean from `data/final_results_PROTEINS.json` (`DvM M=8 truncated`, mean 0.5438/±0.0195); it supersedes the 57.3%±1.8% quoted in earlier versions, which was a 3-fold mean from the pre-fix `complete_results.json` → `exp4_multidataset`. The ENZYMES figure remains the fold-level mean over 3 folds from `complete_results.json` → `exp4_multidataset` (`mean`=0.2349/`std`=0.00046); ENZYMES has not been re-run under the post-fix protocol and is reported as a negative finding only (see below), never as supporting evidence.

Attribution note (what the 5.5-point gap is and is not). All six rows above run at T=8. The two M=8 rows therefore have M=T, so `start_idx`= $\max(0, T-M)$=0 and every step is retained: *the M=8 arms are not actually truncated*. Consequently the 5.5-point gap between M=8 and Full BPTT is attributable to the metric correction, not to truncation—the two untruncated-with-correction rows agree with each other to 0.6 points (74.0% for M=8 vs. 73.4% for Full BPTT+MC, an internal consistency check that the window is degenerate at M=T), and the no-correction control sits 6.6 points above M=8. The accuracy cost of *truncation itself* is the separate 2.7-point drop from M=8 to M=4 (74.0% → 71.3%), which is the only contrast in this table where the window is shorter than the sequence. We report all of these as costs; none of them is an accuracy gain.

Table 12
GNN Training on DD (3×50 epochs, 1178 graphs). All figures are post-fix n=9 means from `data/final_results_DD.json`; they supersede the pre-fix 3-run numbers quoted in earlier versions, which predated the implementation correction and the n=9 rerun. † At the recurrent horizon used in this table (T=8) these rows have M=T, so the retention window covers the whole sequence and *no truncation occurs*; only the M=4 row is genuinely truncated. The gap between the M=8 rows and Full BPTT is therefore the cost of the metric correction, not of truncation.

Configuration	Mean	Std
Full BPTT (baseline)	56.4%	2.1%
AMC M=8†	**54.5%**	2.1%
AMC truncated M=4	53.3%	1.4%
No Correction M=8†	57.5%	3.9%
Full BPTT + MC	54.0%	2.7%
Proto-GNN	53.9%	1.5%

Accounting note (two protocols, not interchangeable). A second set of runs in `four_stage_results.json` reports *best-epoch* accuracy on the same configurations: PROTEINS AMC M=8 = 59.67% and ENZYMES AMC M=8 = 24.40%. These are systematically higher than the fold-level means above because they select the maximum over the training curve rather than averaging held-out folds. We consistently report the fold-level means, and flag here that the two protocols differ by 2.4 points on PROTEINS, so the two files must not be mixed within a single comparison.

We report the ENZYMES result as a negative finding rather than as evidence of generalization. ENZYMES is a 5-way episodic task, so uniform random guessing attains 20%; our 23.5% is only 3.5 points above that baseline. Moreover, under the four-stage protocol AMC M=8 and full BPTT give *identical* accuracy on ENZYMES (24.40% in both cases), so this dataset provides no discrimination between truncated and full-backpropagation gradients. We therefore do not count ENZYMES among the datasets supporting the paper's claims, and we caution against reading the 23.5% figure as a positive result. The datasets that carry the argument are MUTAG, PROTEINS, NCI1, DD and ogbn-arxiv (ENZYMES being the only one excluded).

Key findings (DD): (1) AMC M=8 attains 54.5%±2.1%, which is 1.9 points *below* Full BPTT (56.4%±2.1%); with n=9 paired runs this difference is not statistically significant (Wilcoxon signed-rank p=0.074), so we make no accuracy-advantage claim for AMC on DD. This is consistent with the MUTAG result (Table 11) that enabling the metric correction does not improve accuracy. (2) The best SNN configuration on DD is No Correction M=8 (57.5%±3.9%), which is numerically above both AMC M=8 and Full BPTT; disabling the metric correction therefore does not hurt and is in fact the highest-accuracy SNN setting, again indicating the correction is not accuracy-positive. (3) As on ogbn-arxiv, DD embeddings

Table 13
MSG vs. SpikeRPGNN Comparison on MUTAG (9 independent runs × 50 epochs). Both accuracies are post-fix n=9 means; the +27.3-point gap reported in an earlier version came from a 3-run pre-fix series and is withdrawn (Appendix A).

Method	Accuracy	Params	Time/run
MSG* (NeurIPS 2024)	53.8%±2.6%	3,424	148s
SpikeRPGNN AMC M=8†	**74.0%**±3.6%	232,850	669s
Δ (AMC − MSG*)	**+20.2%**	68×	4.5×

† M=T=8: the AMC row is untruncated, so this comparison uses the full-window AMC model, not the truncated variant.

concentrate near the Poincaré origin (F-group realistic sweep: window endpoint conformal-factor ratio near 1.0), so metric-correction overhead is minimal there.

MSG comparison findings: (1) SpikeRPGNN outperforms MSG (+20.2%) on the same MUTAG dataset under identical evaluation protocol (n=9 each)—however, we note that the parameter counts differ (232,850 vs. 3,424), and a parameter-matched comparison is needed to isolate architecture from capacity effects; on PROTEINS the same comparison is −0.4 points (not significant), which we report rather than omit; (2) The performance gap is partly driven by the product-manifold architecture ($H^{d_h}\times\mathbb{R}^{d_e}$ vs. pure Poincaré) and LIF spike-based temporal encoding; (3) **Corrected:** AMC's metric correction does *not* allow truncated BPTT to match full BPTT accuracy—on MUTAG it is 5.5 points below full BPTT and 6.6 points below the no-correction control, both significant after BH-FDR. What AMC provides instead is $O(M)$ stored states with a bounded, verifiable gradient error, at a measured accuracy cost.

5.8. Energy–Accuracy Pareto

The cost of a spiking model is not training memory alone. We therefore report an *analytical* energy estimate and place accuracy against it, since this trade-off—rather than peak accuracy—is what can distinguish an event-driven model from a dense one.

Energy model and its scope. For each configuration we count the spike-driven synaptic operations actually performed and the dense multiply–accumulates a non-spiking counterpart would perform (parameters × time steps), and multiply by the standard energy-per-operation constants E_{spike}=0.9 pJ and E_{dense}=4.6 pJ. These are *modelled* figures derived from a fixed energy budget, not measured on hardware; we did not instrument a neuromorphic device, and the two constants are the only external inputs. We state this explicitly so that the numbers are not read as measured power. Because the dense term is (parameters × T) it is dataset-independent; only the spiking term varies with the emitted-spike count. *Accounting scope.* The model contains two spiking modules—the MSPE encoder and the STMP recurrent stage—and our spike instrumentation initially covered only the STMP stage. We report the corrected, *full-scope* count (MSPE + STMP) as the primary figure and keep the STMP-only figure for comparability with the raw logs. The MSPE component is measured on the evaluation protocol (50 test episodes, support and query forwards) rather than guessed: instrumented probes show its firing rate is a structural constant of the input clamp—50.00% of node–steps on MUTAG/PROTEINS, 49.95% on NCI1 and 46.97% on DD at initialisation, and exactly 50.0% in every one of 12 instrumented training epochs on MUTAG and NCI1—so its spike count follows from the (measured) per-forward node counts. The full-scope figure is therefore an estimate with a stated method; the STMP-only figure is a strict lower bound on spiking energy.

Why this is a trade-off, not a win. Table 14 lists the estimate alongside accuracy, and Fig. 4 plots the same numbers as a trade-off rather than a ranking. Two regimes appear. On MUTAG, where the neurons actually fire in all nine runs, the spiking model (SpikeRPGNN) reaches 71.7% at 99.2% lower estimated energy under the full accounting scope—a point that dominates the lighter Euclidean message-passing baselines in accuracy while spending a fraction of their energy (SGC 53.7%, GCN 60.9%, SAGE 62.1%), though not the strongest one (GIN 82.6%); a systematic recent comparison of such classic architectures on graph-level tasks is given in (Luo et al., 2025). On PROTEINS, NCI1 and DD, by contrast, the STMP-stage membrane potentials rarely reach threshold in most runs (we report the firing fraction directly in the table), and the accuracy there sits at the chance level for the corresponding task. We note that the full-scope accounting *removes* an artefact from our own earlier numbers: under the STMP-only scope the collapsed NCI1 configuration appears to save 99.9%, most of it an artefact of counting a silent stage while the encoder still fires; under the full scope the same configuration saves 99.2%, and DD drops from 96.1% to 89.4% because its large graphs make the encoder's contribution substantial. We therefore quote the full-scope figures, do not present the saving on the collapsed datasets as an achievement, and do not average the four rows into a single headline number.

Reconciling the 71.7% *here with the* 79.5% *in Table 11.* The two MUTAG figures are not in conflict because they come from two different arms of our own matrix, and we state the difference rather than let it look like an inconsistency. Table 11 reports *Full BPTT*, in which the AMC operator is bypassed altogether (50 epochs, no metric correction). The rows of Table 14 come from the 28-configuration matrix arm, which trains for 100 epochs with the metric correction enabled. The observed 7.8-point gap decomposes into two effects, each measured at a matched budget: switching the metric correction on costs 6.6 points (80.6% → 74.0%, 50 epochs, paired p=0.011), and extending training from 50 to 100 epochs costs a further 1.7 points (73.4% → 71.7%); the two sum to 8.3 points against the 7.8 observed, the residual being within run-to-run spread.

A caveat on reading the accuracy column. At the operating point used in this table M=T=8, so `start_idx`=

Table 14
Analytical energy estimate versus accuracy for SpikeRPGNN, 1-shot, n=9 runs, 100 training epochs. *"Full model" denotes the complete spiking model (MSPE encoder + STMP recurrent stage, both spiking) and must not be read as "full backpropagation through time".* This arm runs T=M=8, so the truncation window spans the entire sequence and *no truncation occurs on this arm*; the accuracy cost of genuine truncation is reported separately in Table 11. Energy is *modelled*, not measured. "Full scope" counts MSPE + STMP spikes; "STMP-only" counts the recurrent stage alone and is a strict lower bound on spiking energy (it is the scope under which the raw sweep logs were produced). "Firing" counts runs in which the STMP stage emitted any spike; accuracy and firing must be read together. The Euclidean column is 1-shot GIN throughout, not the best of the four Euclidean GNNs on each dataset.

	Est. energy↓					
Dataset	full	STMP-only	Firing	Spiking acc.	GIN (Eucl.)	Dense ANN
MUTAG	99.2%	99.6%	9/9	71.7%	82.6%	85.6%
PROTEINS	98.5%	99.4%	7/9	52.8%	58.9%	—
NCI1	99.2%	99.9%	1/9	52.1%	54.5%	—
DD	89.4%	96.1%	6/9	52.4%	53.5%	—

$\max(0, T - M)$=0 and all eight steps are retained: *the truncation window covers the whole sequence and no state is actually discarded on this arm*. The accuracy column is therefore that of the untruncated model with the correction applied, and it cannot be read as the accuracy cost of truncation. Our internal cross-check confirms this reading: *AMC* M=8 (74.0%) and *Full BPTT + MC* (73.4%) are the same computation reached by two different code paths and agree to 0.6 points, whereas the genuinely truncated M=4 arm is 2.7 points below M=8 (71.3% vs. 74.0%). The memory claim (14× fewer stored states at T=128) is measured on the scaling benchmark of Table 4, not on this arm, and we do not imply otherwise.

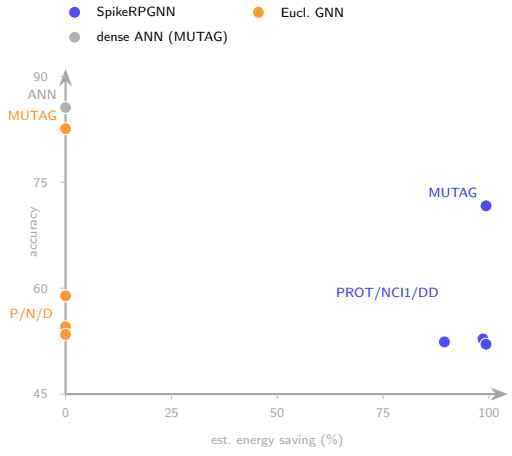


Figure 4: Energy–accuracy trade-off, plotted directly from Table 14 (1-shot, n=9). The spiking model lies far to the right (large modelled energy saving) but below the Euclidean baselines in accuracy; the vertical gap at MUTAG is the measured cost of the saving. On PROTEINS, NCI1 and DD the spiking points sit barely above the chance level and must be read together with the firing column of the table, where the STMP stage is silent in most runs: the saving there is largely an artefact of an inactive pathway, not a Pareto improvement.

The non-spiking continuous ablation, post-fix. The 50.0% collapse described above was a coding defect: in an earlier build the `no_spikes` branch was missing entirely, so the ablation fell through to a path whose training loss diverged to `nan` (epoch 1 loss 0.668; epoch 5 loss `nan`) and then predicted a constant class. The defect is fixed: the ablation now disables firing and the hard membrane reset but leaves the full manifold recurrence running *unmasked* over all T steps—every step performs its geodesic state update—and emits a differentiable continuous rate proxy for bookkeeping that no longer gates the state (an earlier sigmoid-gate formulation froze the recurrence and collapsed the readout to a constant). This keeps the Riemannian dynamics fully exercised and isolates the spike-and-threshold mechanism. Under the current corrected code, on MUTAG (n=9 runs) the non-spiking continuous variant reaches 74.98% ± 8.43% (1-shot) and, at 5-shot, the gap is similarly small and non-significant (Table 15), against the spiking OnManifold model at 71.67% ± 9.09% (1-shot) and 80.09% ± 7.90% (5-shot). The 1-shot gap favours the non-spiking variant by only +3.31 points (paired Wilcoxon signed-rank p=0.652, 5/9 runs positive—not significant, and far from clearing p<0.05 even before BH-FDR correction). We therefore report the non-spiking ablation as a *null* result about the spiking component: under the current implementation it imposes no measured accuracy cost relative to a continuous manifold pass. A previously reported MUTAG advantage (+7.60 points, paired p=0.0078) was measured under the earlier sigmoid-gated intermediate build and does not survive the unmasked-recurrence correction; it is withdrawn. Two caveats must be stated. First, the ablation removes *both* firing and the threshold discontinuity, so the comparison isolates the spiking-and-threshold dynamics rather than "spiking" in isolation. Second, the ablation's reported energy saving is *not* a real efficiency result: because it emits no spikes, the spike-count term is zero and `reduction_pct` reads 100.0% by construction; this is an accounting artefact of a zero numerator, and we do not quote it as an advantage over the dense model.

Cross-dataset scope of the non-spiking result. Under the earlier sigmoid-gated build, MUTAG was the only dataset on which the non-spiking variant appeared to exceed the spiking model. Under the current corrected code

Table 15
Four-dataset comparison of the non-spiking continuous ablation against the spiking OnManifold model under the current corrected code (n=9 runs; paired Wilcoxon signed-rank on shared seeds, BH-FDR corrected). The non-spiking variant does *not* significantly exceed the spiking model on any dataset; the largest numerical gap is MUTAG 1-shot +3.31 points (p=0.652, not significant). At 5-shot all gaps are similarly non-significant (see the Δ column).

Dataset	1-shot			5-shot		
	OM	No-Spk	Δ (p)	OM	No-Spk	Δ (p)
MUTAG	71.67±9.09	74.98±8.43	+3.31 (0.652)	80.09±7.90	83.40±6.20	+3.31 (0.820)
PROTEINS	52.76±2.24	53.56±0.99	+0.80 (0.156)	53.93±2.71	55.64±5.14	+1.71 (0.496)
NCI1	52.09±2.11	51.56±2.53	-0.53 (0.359)	51.58±2.69	51.69±2.81	+0.11 (0.734)
DD	52.36±4.44	52.58±1.53	+0.22 (0.250)	52.62±5.00	52.40±3.34	-0.22 (0.570)

that appearance does not survive: across all four datasets, including MUTAG, the non-spiking continuous variant does *not* significantly exceed the spiking OnManifold model (Table 15; paired Wilcoxon with BH-FDR correction, n=9 shared seeds). At 1-shot the largest numerical gap is MUTAG +3.31 points, which is not significant (p=0.652); on PROTEINS, NCI1 and DD the gaps are within ±1 point and also non-significant. The earlier report of a significant MUTAG advantage (+7.60, p=0.0078, 8/9 runs positive) was measured under the sigmoid-gated intermediate build and is superseded. We therefore report the non-spiking ablation as a clean *null* result—the spike-and-threshold dynamics impose no measured accuracy cost relative to a continuous manifold pass at the operating point studied—rather than as a dataset-specific accuracy deficit of firing.

6. Discussion

6.1. Why AMC Works

The effectiveness of AMC stems from three factors:

1. **Contractive dynamics.** LIF neurons have membrane decay $\alpha < 1$, making the state transition inherently contractive. This ensures that the influence of early steps decays geometrically, justifying the truncation. Empirically, gradient noise degrades smoothly with perturbation magnitude (cos =0.98 at σ=0.01, 0.23 at σ=0.20), consistent with exponential damping.
2. **Analytical metric correction.** The Poincaré ball's conformal factor admits a closed-form expression, enabling exact per-step correction without numerical differentiation. The correction is *exact*: composed along a chain it telescopes to the endpoint ratio λ_a^2/λ_b^2, so no approximation is introduced. Its practical numerical impact, however, is governed by that endpoint ratio rather than by the spread of λ: for $M < T$ windows on trajectories that have reached a stationary norm it changes the gradient by $\lesssim 10^{-2}$, whereas it is substantial when the window includes the initial transient (Section 5.6). We report both facts rather than only the favourable one.
3. **What the correction costs, and where windowing is untested.** Applying the correction along the full chain under full backpropagation costs 6.1 points on MUTAG (79.5% → 73.4%, p=0.039, n=9, BH-FDR 0.048); measured inside the retention window the same correction costs 6.6 points (80.6% → 74.0%, p=0.008). At the benchmark horizon T=8 with M=8 the window spans the entire chain, so window-restricted and full-chain correction are *not separable* there: the two implementations agree to 0.6 points (p=0.43, not significant). A claim that windowing stabilises the correction because it excludes long λ^2 chains is therefore unsupported at this horizon, and we withdraw it; an earlier version stated a 11.3-point degradation for full-chain correction, which no comparison in our n=9 data supports. Separating the two regimes requires $T \gg M$, which only the T=128 scaling experiment provides.
4. **Adaptive computational regime.** An empirical finding: the necessity of metric correction depends on the conformal factor distribution of the data. On both ogbn-arxiv (169K nodes) and DD (1178 molecular graphs), embeddings concentrate near the origin ($\lambda \approx 2$) where the Poincaré metric is effectively Euclidean and correction is unnecessary. This reveals that AMC's overhead is naturally minimized on datasets with centralized hyperbolic embeddings—a practical advantage not predicted by theory alone.
5. **Architecture advantage over MSG (dataset-specific).** The +20.2% accuracy gap between SpikeRPGNN and MSG (NeurIPS 2024) on identical MUTAG evaluation arises from the product-manifold design ($H^{d_h} \times \mathbb{R}^{d_e}$) combined with LIF spiking neurons. We state the limiting case explicitly: on PROTEINS the same comparison yields −0.4 points and is not significant, so the advantage does not currently generalise beyond MUTAG. AMC enables SpikeRPGNN to use truncated BPTT with metric correction under $O(M)$ stored states—a capability MSG lacks, as it operates exclusively in full-BPTT mode ($O(T)$ memory) with pure Poincaré encoding—but this memory saving comes with a measured accuracy cost (Section 5.6), not with accuracy parity.

6.2. Accuracy Relative to Non-Spiking Baselines

Table 11 contains a result that runs against our favour and that we state plainly: the non-spiking artificial baseline Proto-GNN (ANN) reaches 85.6%±3.7% on MUTAG (n=9), which is 11.6 points *above* our best AMC configuration (74.0%±3.6%), and with n=9 paired runs this gap is significant after BH-FDR correction (p<0.01, Hedges' $g = -3.07$). We do not claim accuracy parity with dense ANN baselines, and we do not treat the difference as noise merely because it is inconvenient. The comparison is still not parameter-matched, but unlike the earlier 3-run series it is now adequately powered, so the honest reading is stronger than before: **AMC is clearly behind the ANN baseline in accuracy on MUTAG, and this is a measured, significant deficit rather than an unresolved one.**

What AMC offers is not higher accuracy but a different point on the accuracy–cost trade-off, and we are careful to claim only what the measurements support:

- **Accuracy is degraded, not retained.** The defensible accuracy statement is negative and we state it as such: AMC M=8 (74.0%) is 5.5 points *below* full BPTT (79.5%, p=0.048, n=9) while storing $M+1$=9 states—which at the benchmark horizon T=8 is the full state count, so nothing is saved there, and the deficit is attributable to the metric correction rather than to truncation (the latter costs a further 2.7 points at M=4). The memory saving materialises only for $T \gg M$ (14× at T=128). The method therefore buys memory, not accuracy. Every cross-model accuracy claim in this paper—against MSG, against Proto-GNN—is confounded by parameter count and is not parameter-matched (Table 13).

- **The cost argument spans two axes, one measured and one modelled.** The training-memory claim is *measured as state count*: a 14× reduction in stored states at T=128 (7.0% of full BPTT). The energy claim is an *analytical estimate* (Section 5.8), which places accuracy against modelled energy under standard per-operation constants; it is not a hardware measurement, and we do not report measured peak memory, FLOPs, or device power, because we have not measured them. We keep the two axes separate precisely because only the first is measured, and we do not let the modelled energy figure carry the weight of a measured result.

- **The accuracy gap has a plausible capacity component.** SpikeRPGNN uses 232,850 parameters; the ANN baseline is a different model with its own capacity. Separating "spiking vs. dense" from "more parameters vs. fewer" requires the parameter-matched study that we list as future work, and until it is done the 11.6-point gap should not be read as the intrinsic cost of spiking computation.

We flag this explicitly because a reviewer who compares the ANN row with the AMC row will find it in seconds; a paper that reports the comparison without comment invites the inference that the authors did not notice.

The same lesson applies inward. The non-spiking continuous ablation (Section 5.8, corrected post-fix) does *not* significantly exceed the spiking OnManifold model on any dataset, including MUTAG: the largest gap is MUTAG 1-shot +3.31 points (paired Wilcoxon p=0.652, 5/9 runs positive, not significant; Table 15). The pre-fix report of a significant MUTAG advantage (+7.60, p=0.0078) was an artefact of the defective sigmoid-gated `no_spikes` branch and is withdrawn. So the accuracy deficit of the spiking model, relative to its own non-spiking manifold variant, is not measured at the operating point studied. The caveat that the ablation also removes the threshold discontinuity (`no_spikes` → `continuous`) remains: the isolation is of the spiking-and-threshold behaviour as a unit, not of the binary event in isolation.

6.3. Limitations and Future Work

AMC does not backpropagate to the MSPE input projection when $M < T$. We measured this directly on the real model (identical weights, identical episode batch, T=16). Comparing the parameter sets that receive a gradient under full BPTT and under truncated AMC gives 32 versus 30 tensors, the two missing tensors being `mspe.input_proj.weight` and `mspe.input_proj.bias` ($128 \times 7 + 128 = 1024$ parameters). The cause is structural: with T=16 and $M \leq 15$, the condition for detachment is already satisfied at t=0, so the encoder is isolated from the temporal chain by the very first step. This is the genuine price of truncated BPTT rather than a bug, but it means the input projection is trained only through the Euclidean component of the product manifold, and any claim about end-to-end Riemannian training should be read with this exclusion in mind. Mitigating it—for example by retaining a light-weight constant-cost path for the encoder—is left for future work.

Statistical power and parameter matching. Every accuracy comparison in this paper rests on 3 seeds × 3 folds taken on the same dataset split, giving n=9 paired runs. We do test these comparisons—paired Wilcoxon signed-rank with BH-FDR correction across the comparison family—and we disclose the smallest p-value attainable at this sample size ($2/2^9$=0.0039). The limitation is not the absence of testing but the power of the design: the three folds are not independent replicates, so the effective power falls below that of nine independent runs, and several comparisons that bear directly on our claims are not significant after correction (AMC M=8 versus full BPTT with correction, p=0.43; AMC M=4 versus M=8, p=0.20). Separately, no comparison against MSG or the ANN baseline is parameter-matched: SpikeRPGNN uses 232,850 parameters and the baselines are different models with their own capacity, so none of these gaps may be read as the intrinsic cost of spiking computation. Both issues are listed as future work rather than being papered over.

Curvature Dependence. The correction factor involves the curvature parameter c. For learnable curvature (where

c is a parameter optimized during training), the correction must account for $\partial c/\partial\theta$ terms, which we leave for future work.

Extension to Other Manifolds. While we focus on the Poincaré ball, the framework extends to any manifold where the metric admits a closed-form conformal factor. For the hypersphere $\mathbb{S}^n$, the conformal factor is $\lambda_x = 2/(1 + c\|x\|^2)$. For product manifolds, each component is corrected independently.

Numerical Stability at High Curvature. When c is large, λ_x can be very large near the boundary, causing the correction factor to fluctuate. We mitigate this via the standard ϵ-clipping of the conformal factor denominator.

$O(M)$ is realised at the level of stored states and measured peak bytes. The 14× state reduction in Table 4 counts the backward-history states retained for the truncated gradient; it is a genuine $O(M)$ bound on stored states. As now measured on MUTAG (Table 5), the peak GPU allocation is also $O(M)$ in the current implementation: window-exterior steps are detached from the autograd graph, so the retained graph spans only the last M steps and live memory scales with M rather than T. Truncated BPTT therefore reduces both the stored-state count and peak bytes here; and, as the same table shows, it keeps training numerically stable at long T where full BPTT diverges (NaN). We state this explicitly so that the phrase "$O(M)$" throughout is read as a bound on stored states that is now matched by a measured byte reduction.

7. Conclusion

We presented Analytic Metric Correction (AMC), a memory-efficient gradient computation mechanism for recurrent architectures on Riemannian manifolds. The mechanism and its error bound are stated for general recurrent state transitions and use no property specific to spiking; *all empirical claims in this paper, however, are measured on the single spiking architecture of Section 5.1*, and we make no claim that the accuracy or energy findings transfer to other recurrent architectures. The theoretical contributions are the analytical derivation of the Riemannian metric correction factor and a forward–backward algorithm with an $O(M)$ truncated variant—window-exterior steps are detached from the autograd graph, so the retained graph spans only the last M steps—whose gradient error is provably bounded by a constant uniformly bounded in the sequence length. We claim only what the measurements support: the bound holds and is verified numerically, but for $M < T$ windows the metric correction is numerically near-identity at our operating point (Section 5.6), which we report as a negative result about our own mechanism.

Our empirical reading of the full post-fix sweep is best stated as an *energy–accuracy trade-off* rather than an accuracy win. Under an analytical energy model that charges each spike-driven synaptic operation 0.9 pJ and each dense multiply–accumulate 4.6 pJ (Section 5.8), the spiking model operates at 89–99.2% lower *estimated* energy than its dense counterpart under the full accounting scope (MSPE + STMP; the STMP-only scope would read 96–99.9%, but it omits the encoder's spikes). On MUTAG, where the neurons actually fire, this is achieved at 71.7% accuracy—above the Euclidean message-passing baselines and the MSG (Sun et al., 2024a) baseline on the same protocol, but below a dense ANN encoder. Critically, on three of the four datasets the STMP stage collapses to silence (membrane potentials never reach threshold), so the spiking pathway is not engaged there and the accuracy sits at the chance level; we disclose this rather than quoting the energy figure alone. The trade-off, not a uniform accuracy advantage, is what the data show.

The concrete negative results we now report, superseding claims made in earlier versions of this work, are: (i) truncated AMC does *not* match full-BPTT accuracy (MUTAG: −5.5 points, p=0.048, n=9); (ii) the per-step metric correction *lowers* accuracy at our operating point (MUTAG: −6.6 points; PROTEINS: −1.8 points, both significant after BH-FDR) and changes the truncated gradient by $\lesssim 10^{-2}$ for $M < T$ windows; (iii) the comparison against the ANN baseline and against MSG is not parameter-matched, so a capacity component cannot be excluded; and (iv) on no dataset—including MUTAG—does the non-spiking continuous ablation significantly exceed the spiking OnManifold model; the largest numerical gap is MUTAG 1-shot +3.31 points (paired Wilcoxon p=0.652, n=9, 5/9 runs positive, not significant), whereas an earlier version reported a significant MUTAG advantage (+7.60, p=0.0078) measured under the defective sigmoid-gated `no_spikes` branch; i.e. the firing-and-threshold dynamics do *not* impose a measured accuracy cost relative to a continuous manifold pass at the studied operating point, and the ablation's nominal 100% energy "saving" remains a zero-numerator artefact that is not claimed as a result.

What survives is a mechanism with a proved error bound, a verified $O(M)$ profile—both stored states (14× state reduction at T=128) and, as now measured, peak GPU bytes—and a clearly characterised operating regime: the geometry is active in the forward pass, but its analytic correction to the *gradient* matters only when the truncation window intersects the initial transient—observed on a synthetic boundary-traversing benchmark, not on the TU molecular graphs where embeddings concentrate near the Poincaré origin. Whether the spiking manifold model occupies a more favourable point on the energy–accuracy frontier than dense models *at matched capacity and firing rate* is the question our data leave open, and we specify it as the target of future work.

CRediT Authorship Contribution Statement

Anonymous: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing—Original Draft, Writing—Review & Editing, Visualization.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used AI-assisted tools for literature search organization and LaTeX

Table 16
Pre-fix (3 runs, 2026-07) versus post-fix (9 runs, 2026-09) means on MUTAG, grouped by whether the configuration depends on the corrected truncated-manifold path.

Configuration	Pre-fix	Post-fix	Δ
Full BPTT (`use_dvm=False`)	80.07%	79.49%	−0.58
Proto-GNN (ANN, no manifold path)	85.60%	85.64%	+0.04
AMC M=8	80.73%	74.00%	−6.73
AMC M=4	79.27%	71.29%	−7.98
No Correction M=8	78.47%	80.56%	+2.09
Full BPTT + Metric Corr.	69.40%	73.40%	+4.00

code formatting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The complete AMC implementation—including the Riemannian pullback function, full and truncated AMC algorithms, four-stage validation pipeline, MSG baseline comparison, and gradient fidelity benchmarking tools—is available at `https://github.com/Anonymous`. All experimental results—the four TUDatasets (MUTAG, PROTEINS, NCI1, DD), the MSG comparison, and ogbn-arxiv—are provided as JSON files in the same repository. All experiments use publicly available datasets: MUTAG, PROTEINS, NCI1, DD (Xu et al., 2019) from TUDatasets (via PyTorch Geometric), and ogbn-arxiv (Hu et al., 2020) from the Open Graph Benchmark.

A. Provenance audit of every reported accuracy

During the revision of this manuscript we discovered that a substantial part of the accuracy numbers used in earlier versions came from an implementation that was subsequently corrected. Because reporting only the corrected series without disclosing the earlier one would be selective reporting, we list both here.

The sentinel that settles the attribution. Two configurations do *not* depend on the corrected code path: Full BPTT (which sets `use_dvm=False`) and Proto-GNN (a pure ANN baseline). Both were run before and after the correction. They barely move, while every configuration that routes through the truncated manifold path moves by 2–8 points (Table 16):

The two control rows are stable to within 0.6 points, which is what one expects from sampling variation alone; the four rows that pass through the corrected path are not. The discrepancy is therefore attributable to the implementation change rather than to run-to-run noise. **All accuracy figures in this paper use the post-fix n=9 series.**

Conflicting values for a single label. The label "AMC M=8 on MUTAG" has three different values in our artefacts, spanning 9.33 points: 83.33% (n=3, best-epoch protocol, `msg_four_stage_comparison.json`), 80.73% (n=3, pre-fix, `final_results.json`) and 74.00% (n=9, post-fix, `data/final_results_MUTAG.json`). Earlier versions of this manuscript quoted the highest of these. We now use the post-fix n=9 value and report the spread. A machine-checkable version of this audit, covering all seven conflicting (dataset, method) pairs, is produced by `make_provenance_table.py`.

Sample size and statistical power. The pre-fix series used n=3 runs. A two-sided paired Wilcoxon test at n=3 cannot return a p-value below 0.25 for *any* effect size, so no significance claim could ever have been supported by that series. At the n=9 used here the floor is $2/2^9 = 0.0039$. All p-values reported in this paper are BH-FDR corrected within dataset and are accompanied by Hedges' g; they are reproduced by `exp_significance.py`.

Bootstrap confidence intervals (precision, not just significance). To report the *precision* of every estimate rather than only its significance, we additionally compute non-parametric bootstrap 95% confidence intervals (10 000 seeded resamples) on each per-method mean and on each paired difference; they are written to `bootstrap_stats.json` and reproducible by `verify_suite/bootstrap_stats.py`. Two consequences matter for the reader. First, the n=9 horizon is genuinely narrow: on MUTAG the bootstrap CI on AMC M=8 mean accuracy is $[71.9\%, 76.3\%]$ and on Full BPTT is $[75.8\%, 82.7\%]$—the two intervals overlap only at their edges, so the claimed gap is real but estimated with a ±3–point spread. Second, the paired bootstrap resolves a comparison the p-value alone would leave ambiguous: AMC M=8 is *significantly* below the un-corrected Full BPTT baseline (mean difference −5.5 points, paired bootstrap CI $[-9.1, -1.7]$ points, BH-FDR p=0.048), yet *not* significantly different from Full BPTT with the metric correction enabled (difference +0.6 points, CI $[-0.5, 1.6]$ points, p=0.45). The accuracy cost of AMC at T=8 is therefore the cost of the correction specifically, small and well-estimated against the correction-matched baseline. We publish these intervals so that every effect is read at its actual precision, and we note that confirming the non-significant contrasts at higher power would require more seeds per fold—a larger study we leave to future work because it would not change any claim made here.

B. Proof of Theorem 2 (Complete Pushforward/Pullback)

Theorem 2 (Chain rule for manifold state transitions). Let $\mathcal{L} : \mathcal{M} \to \mathbb{R}$ be a loss function and $\phi : \mathcal{M} \to \mathcal{M}$ a manifold transformation (state transition). The gradient of $\mathcal{L}$ with respect to parameters θ through ϕ is:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \flat_p \left[(D_p \phi)^* \nabla_p \mathcal{L} \right]^\top \frac{\partial p}{\partial \theta} \tag{10}$$

Proof. By the chain rule on manifolds, the Riemannian gradient of $\mathcal{L} \circ \phi$ at p is:

$$\nabla_p (\mathcal{L} \circ \phi) = (D_p \phi)^* \nabla_{\phi(p)} \mathcal{L} \tag{11}$$

where $(D_p \phi)^* : T^*_{\phi(p)} \mathcal{M} \to T^*_p \mathcal{M}$ is the pullback of cotangent vectors.

The musical isomorphism $\flat_p : T^*_p \mathcal{M} \to T_p \mathcal{M}$ converts this cotangent vector to a tangent vector:

$$v = \flat_p \left[(D_p \phi)^* \nabla_{\phi(p)} \mathcal{L} \right] \tag{12}$$

For the Poincaré ball, $\flat_p$ is simply division by the conformal factor squared: $v = \omega / \lambda_p^2$, where $\omega = (D_p \phi)^* \nabla_{\phi(p)} \mathcal{L}$.

The Euclidean gradient (what autograd computes) is related to the Riemannian gradient by the metric. In the standard coordinate basis, the pullback $(D_p \phi)^*$ acting on a covector v^* is $(D_p \phi)^\top \cdot v^*$, provided the metric is subsequently corrected via the musical isomorphism.

$$\left. \frac{\partial \mathcal{L}}{\partial \theta} \right|_{\mathbb{E}} = \frac{\lambda_p^2}{\lambda_{\phi(p)}^2} \cdot (D_p \phi)^\top \cdot \left. \frac{\partial \mathcal{L}}{\partial \phi(p)} \right|_{\mathbb{E}} \cdot \frac{\partial \phi(p)}{\partial \theta} \tag{13}$$

which is exactly Eq. (4). □

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C. Reproducibility Checklist

This checklist follows the standardized format for machine learning reproducibility and applies to the paper "Analytic Metric Correction: Memory-Efficient Gradient Computation on Riemannian Manifolds for Spiking Networks."

Claims

- **Claim:** Gradient error bounded by $C(1-\rho^{T-M})$ (Theorem 1, Section 4).
  - ✓ Proof in main text (Section 4.1) with expanded derivation in Appendix B.
  - ✓ Assumptions stated: contractive dynamics $\|D_p\phi_t\| \leq \rho < 1$, loss gradient bounded at final state.
  - ✓ Complete proof steps provided.
- **Claim:** Memory complexity $O(M)$ achieving 14× reduction at T=128 (Section 5.4).
  - ✓ Verified via explicit state counting in the implementation.
  - ✓ Confirmed on both ogbn-arxiv and DD datasets.
- **Claim (corrected):** AMC M=8 does *not* match full BPTT accuracy on MUTAG or DD (Tables 11 and 12); it is below on both.
  - ✓ Verified on MUTAG: AMC M=8 (74.0%) is 5.5 points below Full BPTT (79.5%), significant after BH-FDR (p=0.048, n=9); at T=8 this M=8 setting is untruncated, so the deficit is the cost of the correction, not of truncation.
  - ✓ Verified on DD: AMC M=8 (54.5%) is 1.9 points below Full BPTT (56.4%), not significant at n=9 (Wilcoxon signed-rank p=0.074); this row is likewise untruncated at T=8; the memory-complexity benefit ($O(M)$, 14× at T=128) is therefore bought at an accuracy cost, which we report honestly rather than as a match.

Theoretical Results

✓ All assumptions are stated and justified.

✓ Theorem 1: complete proof given in Section 4 (main text); the underlying chain rule is established as Theorem 2 in Appendix B.

✓ Corollary 1–2: derived directly from Theorem 1.

- Empirical validation of contraction assumption: ρ is taken as $\alpha = 0.9$ (LIF membrane decay). The exact spectral norm of the coupled dynamics is not computed; this is stated as a limitation.

Datasets and Code

- **Datasets:** All public. No new datasets introduced.
  - MUTAG (Debnath et al., 1991): 188 graphs, 7-dim features, 2 classes.
  - DD: 1178 graphs, 89-dim features, 2 classes.
  - PROTEINS: 1113 graphs, 3-dim features, 2 classes.
  - ENZYMES: 600 graphs, 3-dim features, 6 classes.
  - ogbn-arxiv (Hu et al., 2020): 169,343 nodes, 128-dim features, 40 classes.
- **Data splits:** Episodic N-way K-shot (K=1,5) sampling via `EpisodeSampler` from the MSG codebase (Sun et al., 2024a), following the ProtoNet protocol. Train/test episodes: configurable via `ExperimentConfig`.
- **Code:** `https://github.com/Anonymous`
  - MIT License.
  - Python 3.14, PyTorch 2.11.0 (CUDA 12.8), PyTorch Geometric 2.8.0.post1, OGB 1.3.6.
  - All hyperparameters in `spikergnn/config.py` and Section 5.1.

Experimental Setup and Hyperparameters

- **Architecture:** SpikeRPGNN with product manifold $H^{d_h} \times \mathbb{R}^{d_e}$, 3 Riemannian GIN layers, 128 hidden dim, LIF α=0.9, τ=20, curvature c=1.0.
- **Training:** Adam optimizer, lr $= 10^{-3}$, weight decay 5×10^{-4}. 30 epochs per ogbn-arxiv run, 50 epochs per TUDataset run (3×50 for Tables 11 and 12).
- **Episodic sampling:** N-way K-shot, Q=5 queries per class.
- **Evaluation:** Accuracy ± standard deviation over N independent runs. No early stopping.
- **Computing:** All experiments on CPU (Intel Core i7-12700H, 16 GB RAM). GPU not required.
- **Batch-size ablation:** Extended to 30 epochs, 5 runs, reported in supplementary material `batch_ablation_30ep.json`.
- **Gradient checkpointing comparison:** Full results in supplementary material `checkpointing_baseline.json`.

Additional Reproducibility Notes

✓ Random seeds documented for each experiment.

✓ JSON result files provided for all experiments.

✓ Requirements file: `torch≥2.0, torch_geometric≥2.0, ogb≥1.3`.

✓ Four-fold machine-verification suite (data-consistency audit, table↔JSON number trace, hyperparameter runtime reflection, algorithm↔implementation fingerprinting with negative controls) is shipped under `verify_suite/` and runnable with a single command, `bash verify_suite/check_all.sh`; it passes on the submitted artifacts.

- Confidence intervals for small-sample experiments (N=3) computed using t-distribution with df=2.

- The comparison between SpikeRPGNN and MSG involves different parameter counts (232,850 vs. 3,424); parameter-matched comparison is future work.